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Meltzer et al.

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(54) **ELECTRONIC DOCUMENT PRESENTATION
MANAGEMENT SYSTEM**

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(73) Assignee: **SpoonRead Inc.**, Aptos, CA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

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(63) Continuation of application No. 18/807,883, filed on
Aug. 16, 2024, and a continuation of application No.
(Continued)

(51) **Int. Cl.**

G09B 5/12 (2006.01)

G06F 3/0483 (2013.01)

(Continued)

(52) **U.S. Cl.**

CPC **G09B 5/12** (2013.01); **G06F 3/0483**
(2013.01); **G06F 16/38** (2019.01); **G06F**
16/382 (2019.01);

(Continued)

(58) **Field of Classification Search**

None

See application file for complete search history.

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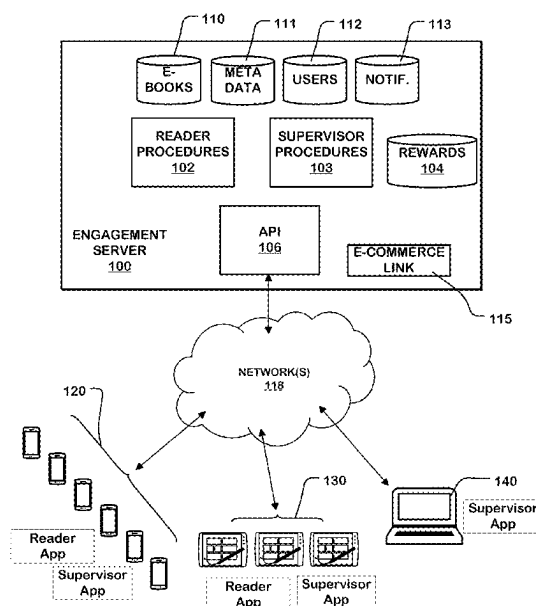
Primary Examiner — Ali H. Cheema

(74) *Attorney, Agent, or Firm* — Ernest J Beffel, Jr.; Paul
A. Durdik; Jason Liao

(57) **ABSTRACT**

A method, system and non-transitory computer readable
storage medium for managing engagement and presentation
of electronic documents is described herein. In an imple-
mentation, the method comprises receiving a registration
request, with a nickname for a child and without personally
identifying information for the child, from a parent account
to set up a nicknamed child account; enabling resource
access via the nicknamed child account on a device; receiv-
ing an e-book request from the parent account to assign one
or more e-books to the nicknamed child account; assigning
the one or more e-books to the device based on the nick-
named child account and authorizing the device to display
content of the one or more e-books to the child; and
receiving and tracking activity data for the child and the
assigned one or more e-books.

20 Claims, 29 Drawing Sheets



Related U.S. Application Data

18/807,891, filed on Aug. 16, 2024, said application No. 18/807,883 is a continuation of application No. 17/671,509, filed on Feb. 14, 2022, now Pat. No. 12,067,900, said application No. 18/807,891 is a continuation of application No. 17/671,505, filed on Feb. 14, 2022, now Pat. No. 12,067,899, said application No. 17/671,509 is a continuation of application No. 17/037,523, filed on Sep. 29, 2020, now Pat. No. 11,250,718, said application No. 17/671,505 is a continuation of application No. 16/685,685, filed on Nov. 15, 2019, now Pat. No. 11,250,717, which is a continuation of application No. 15/798,248, filed on Oct. 30, 2017, now Pat. No. 10,796,591, said application No. 17/037,523 is a continuation of application No. 15/798,248, filed on Oct. 30, 2017, now Pat. No. 10,796,591.

- (60) Provisional application No. 62/569,285, filed on Oct. 6, 2017, provisional application No. 62/484,275, filed on Apr. 11, 2017.

(51) Int. Cl.

G06F 16/38 (2019.01)
G06F 16/48 (2019.01)
G06F 21/31 (2013.01)
G06F 40/106 (2020.01)
G06F 40/123 (2020.01)
G06F 40/131 (2020.01)
G06F 40/143 (2020.01)
G09B 5/02 (2006.01)
G09B 5/06 (2006.01)
G09B 17/04 (2006.01)
G06F 21/62 (2013.01)

(52) U.S. Cl.

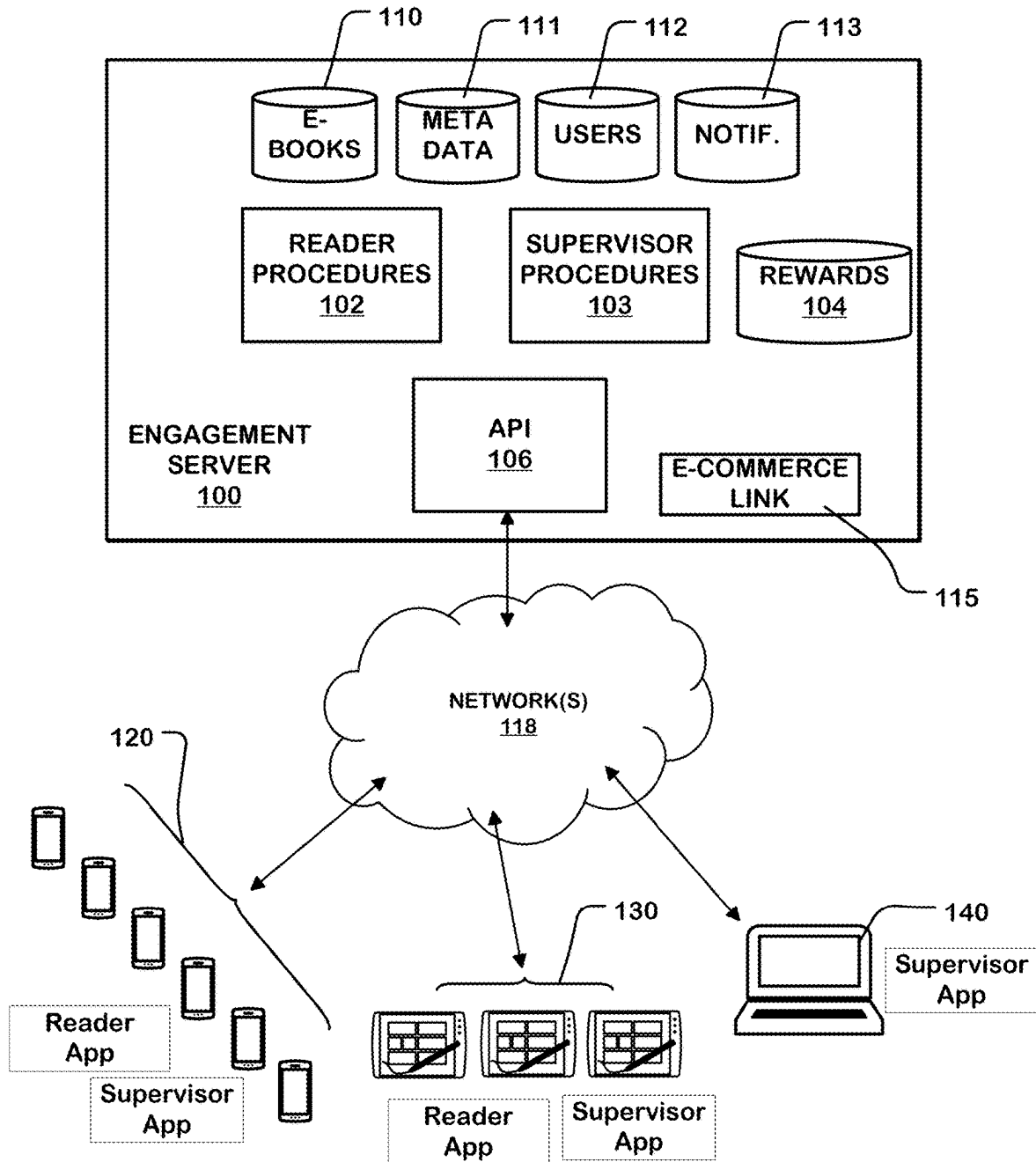
CPC **G06F 16/48** (2019.01); **G06F 21/31** (2013.01); **G06F 40/106** (2020.01); **G06F 40/123** (2020.01); **G06F 40/131** (2020.01); **G06F 40/143** (2020.01); **G09B 5/02** (2013.01); **G09B 5/062** (2013.01); **G09B 17/04** (2013.01); **G06F 21/6218** (2013.01)

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**Fig. 1**

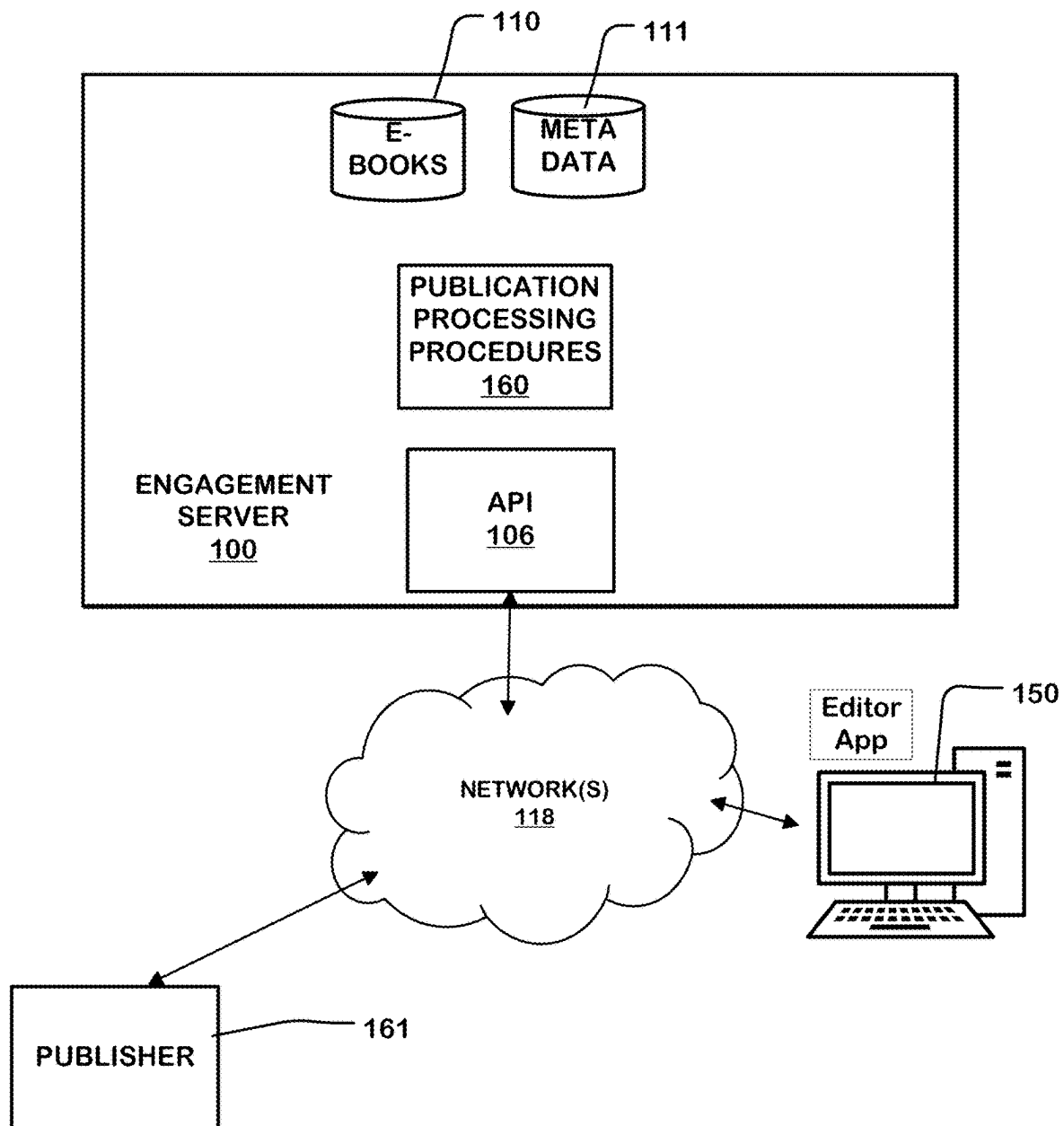
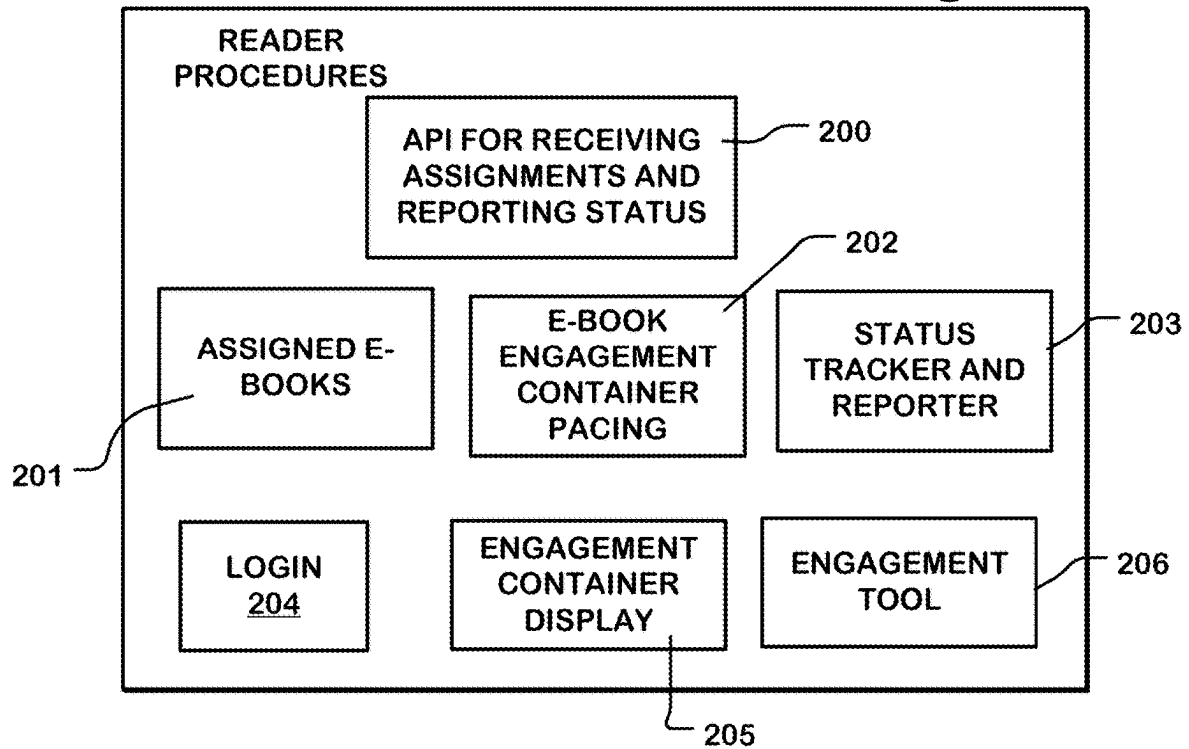
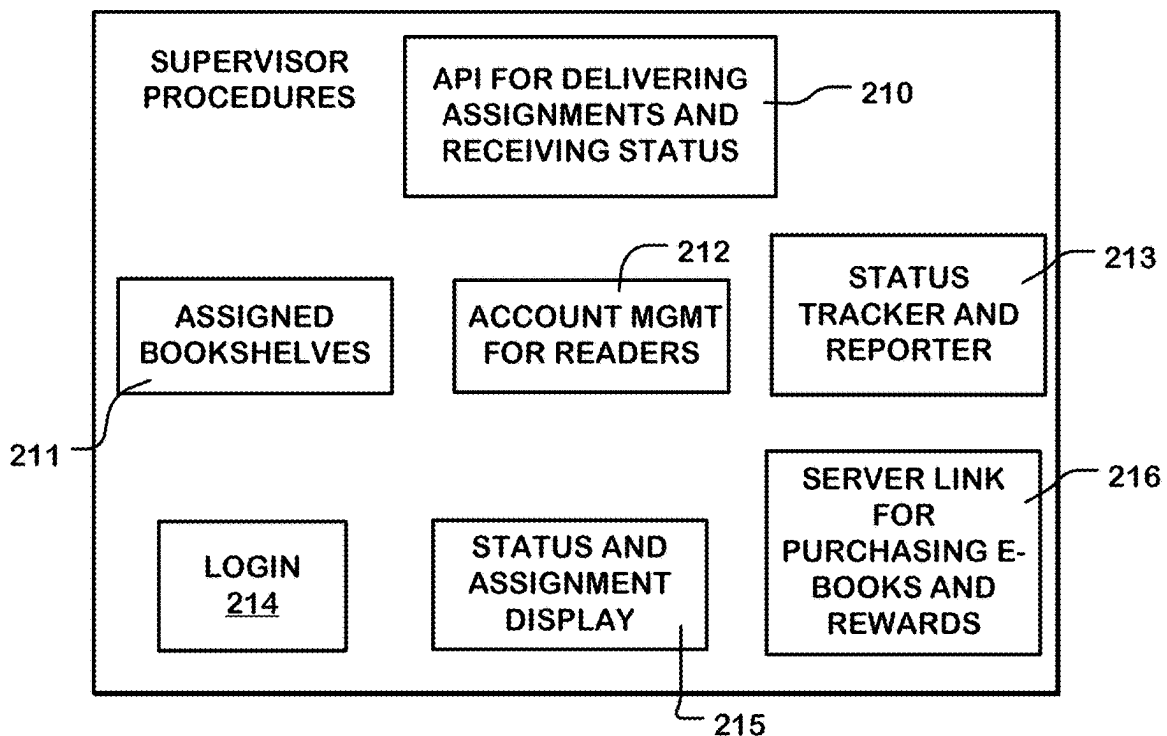


Fig. 2

Fig. 3**Fig. 4**

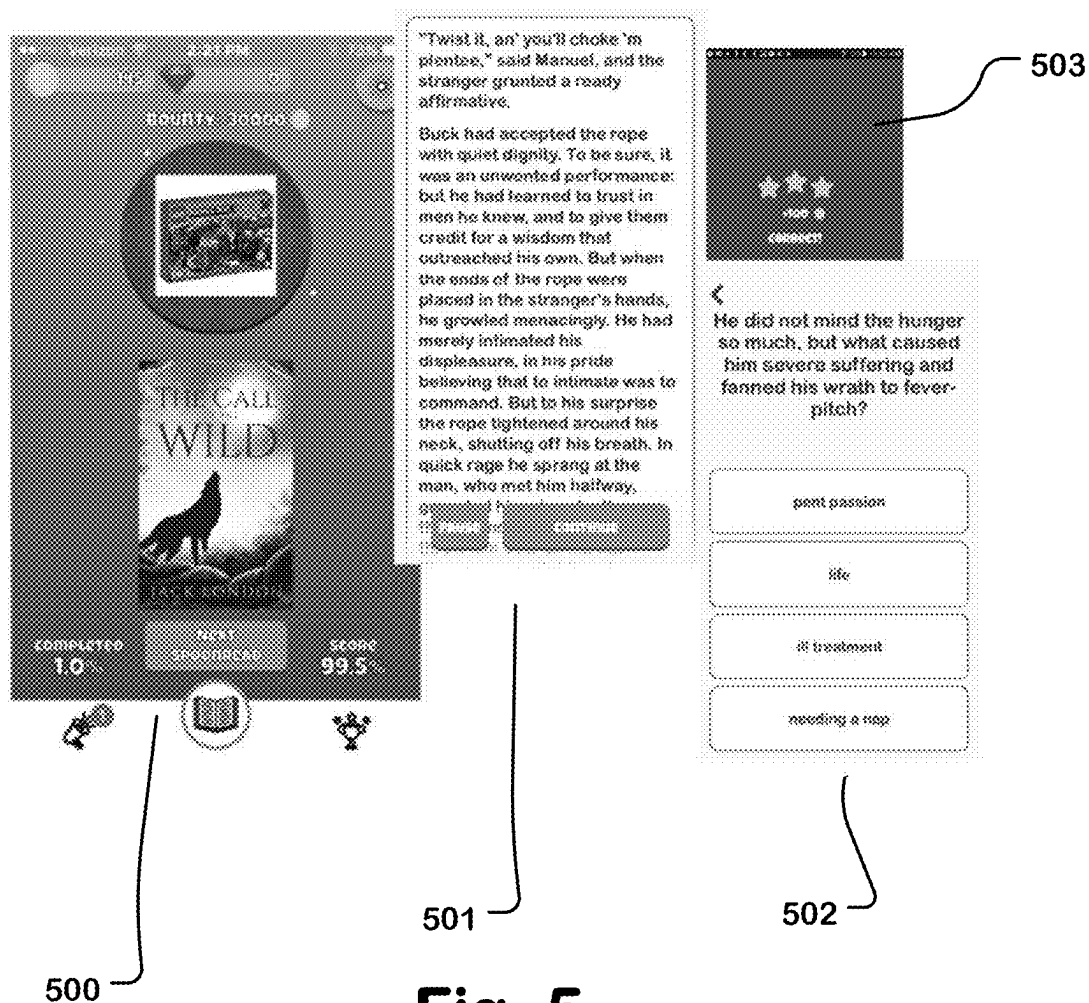


Fig. 5

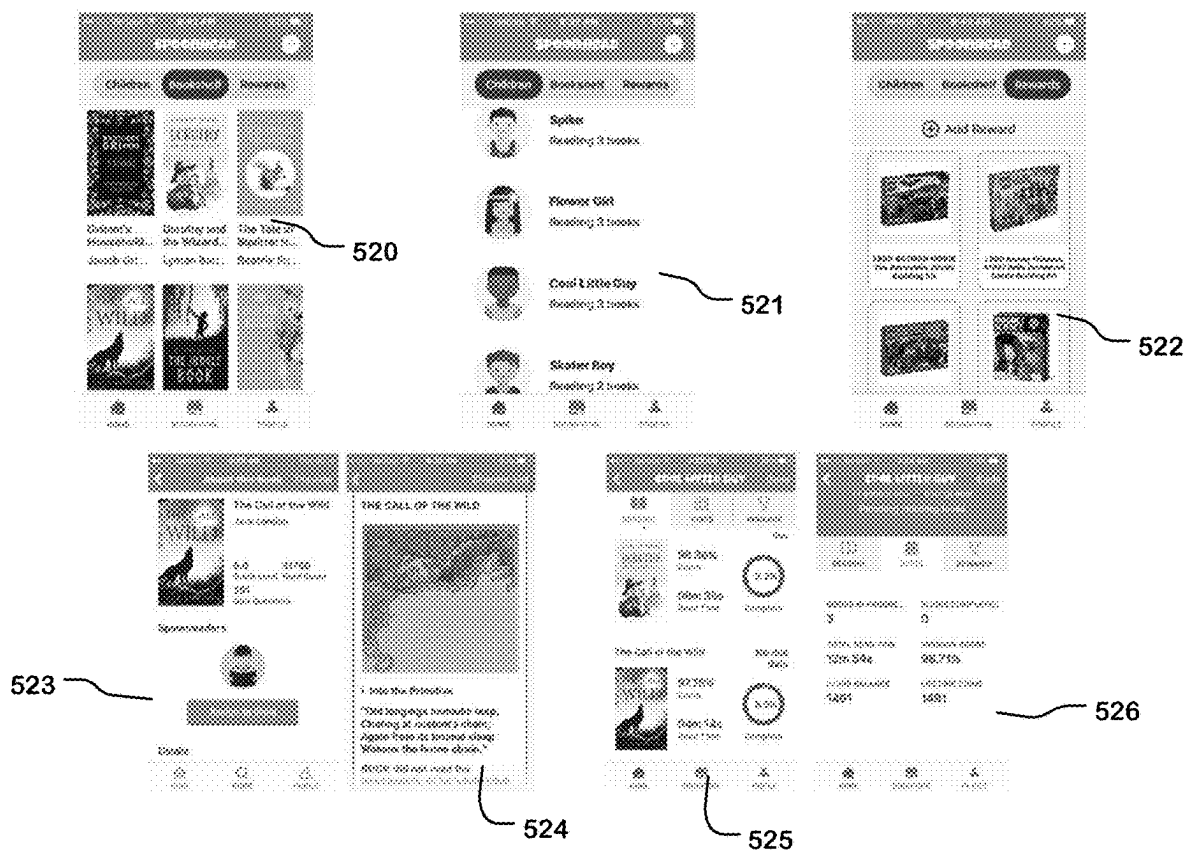
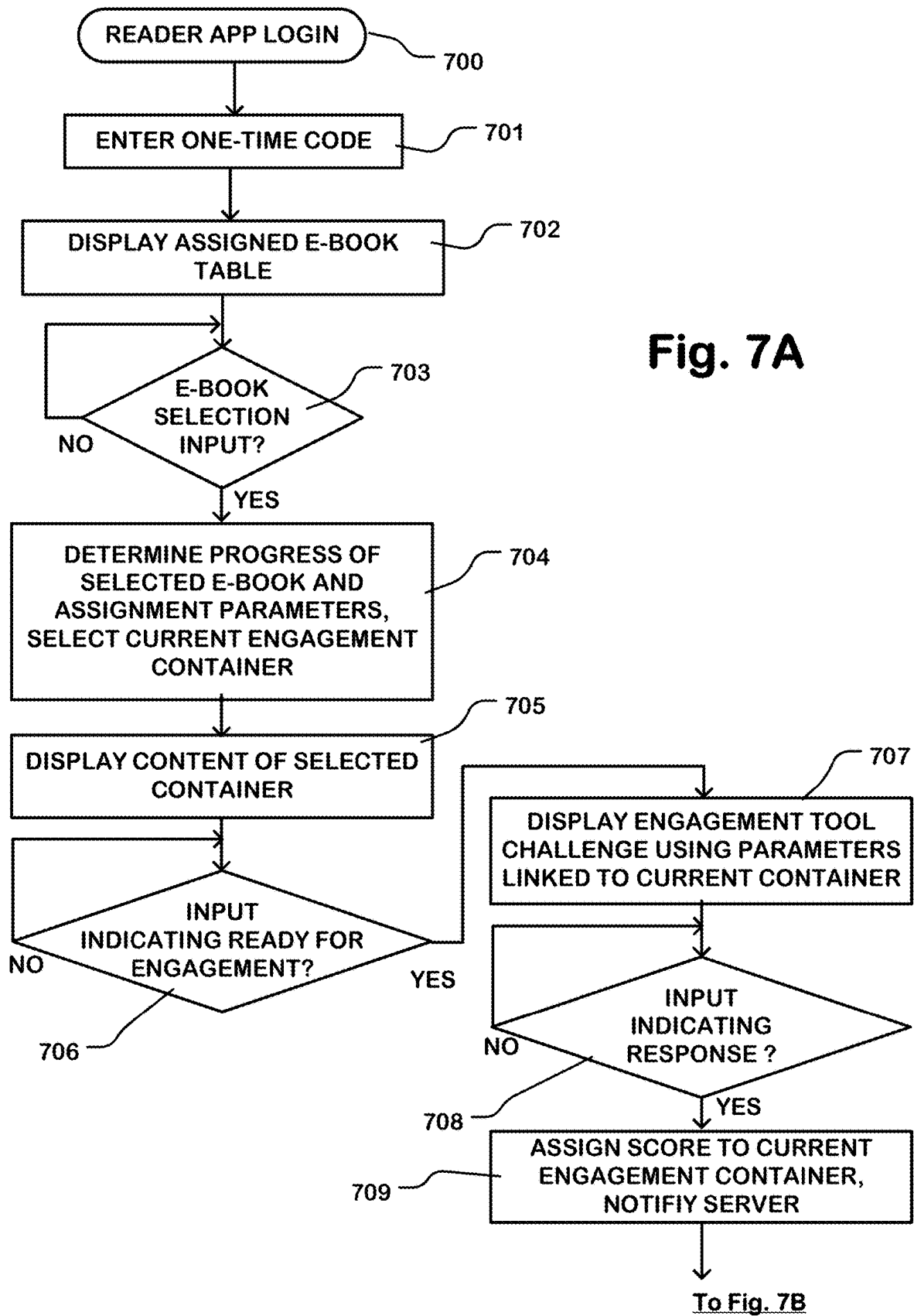


Fig. 6



From Fig. 7A

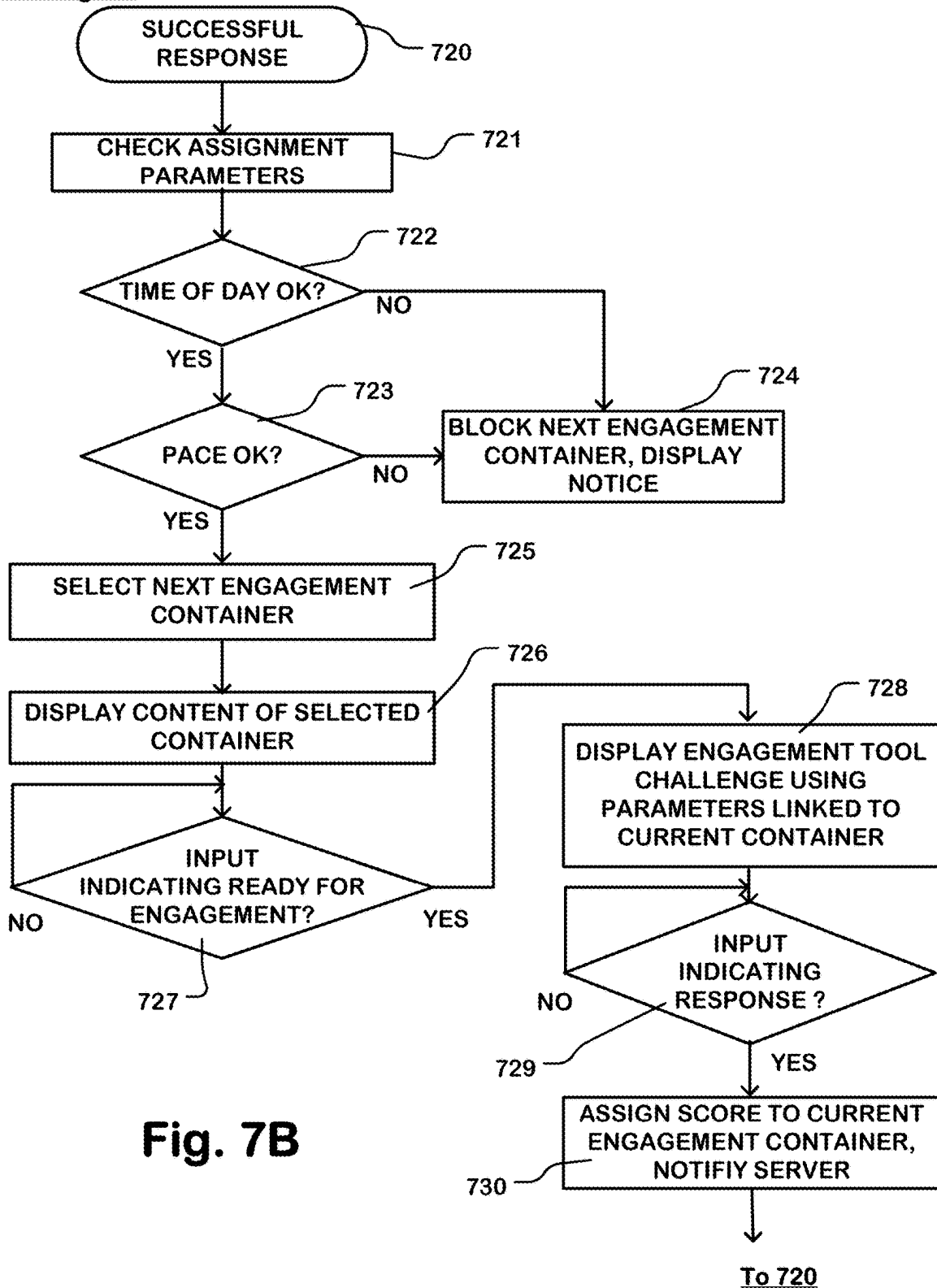
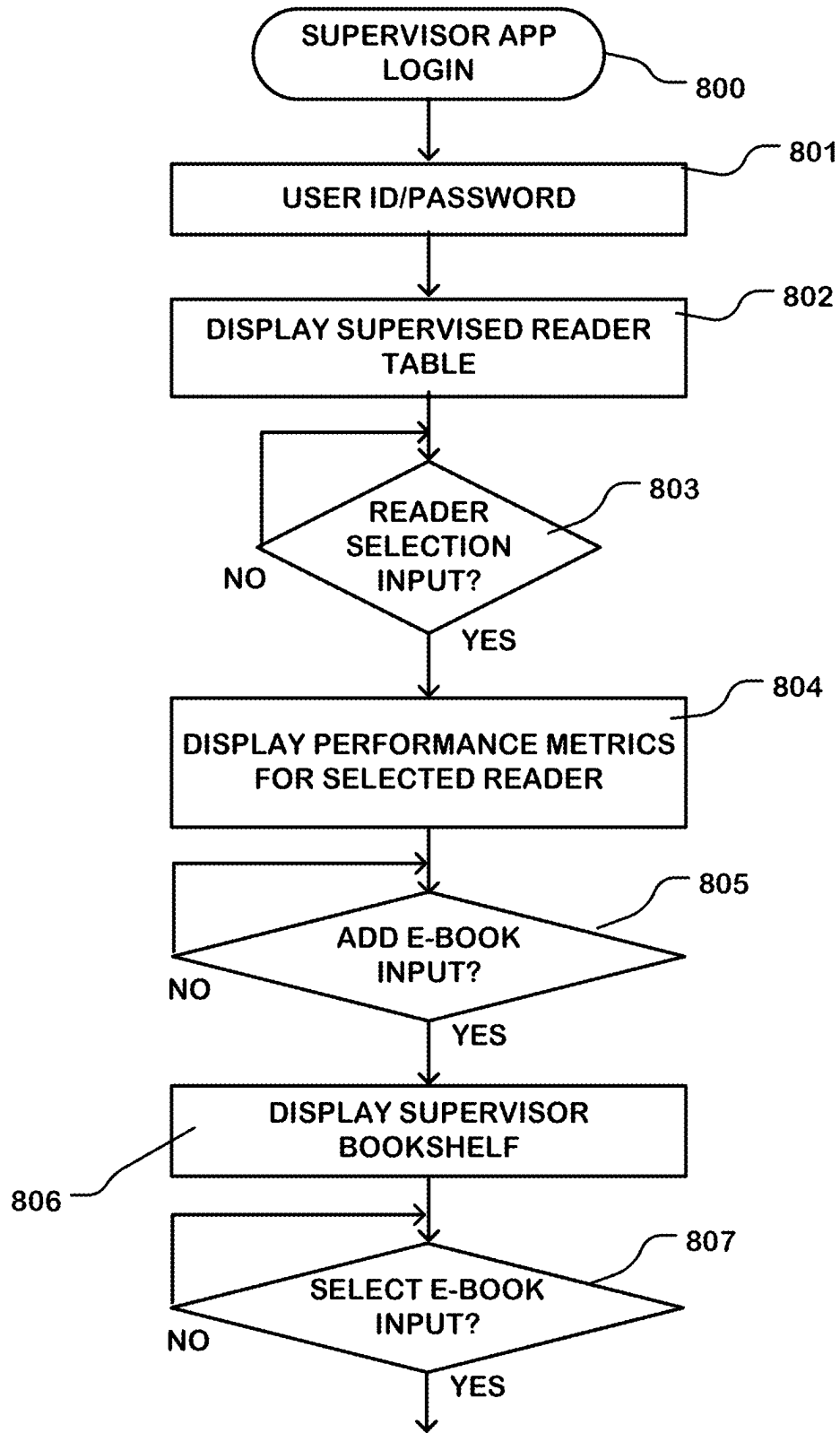
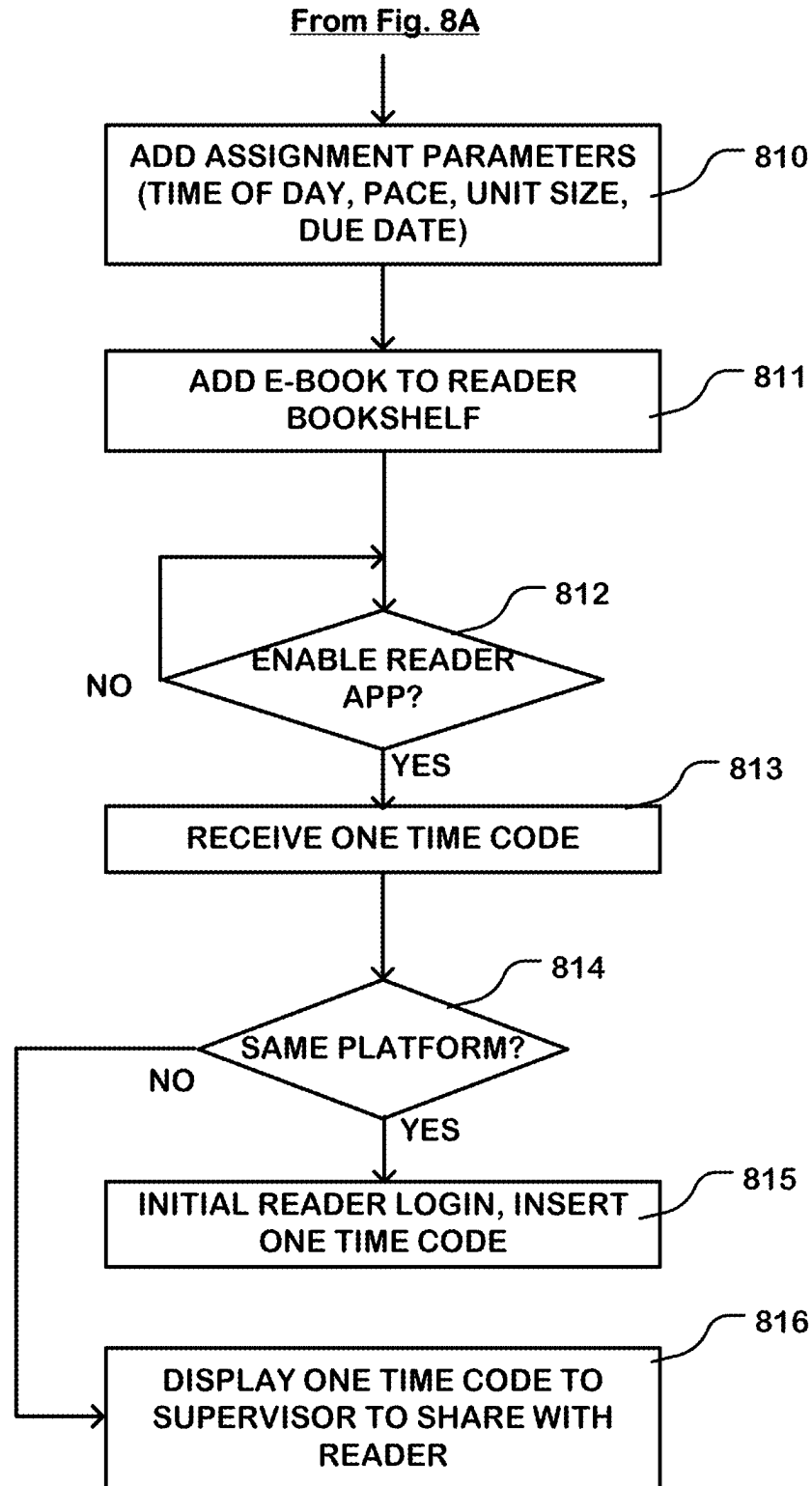


Fig. 7B

**Fig. 8A**

**Fig. 8B**

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<SpoonreadBody>
  <ChapterNumber>01</ChapterNumber>
  <ChapterName>Chapter 1: The Earthquake</ChapterName>
  <Spoonful id="1">
    <p>The train from 'Frisco was very late. It ... stop the conductor called out
in a loud voice:</p>
    <p>"Hugson's Siding!"</p>
    <p>At once a little girl rose from her seat and walkeD...
and an accident happen to his passengers. So he moved the cars slowly and with
caution.</p>
    <p>The little girl stood still to watch until the train had disappeared
around a curve; then she turned to see where she was.</p>
  </Spoonful>
  <Spoonful id="2">
    <p>The shed a...
asleep.</p>
    <p>She set down the bird-cage and poked the boy with her ... eyes briskly.</
p>
  </Spoonful>
  <Spoonful id="3">
    <p>"Hello!" he said, seeing her, "are you Dorothy Gale?"</p>
    <p>"Yes," she answered, looking gravely at his tousled hair a... Ranch?"</p>
    <p>"Of course," he answered. "Train in?"</p>
    <p>"I couldn't be here if it wasn't," she said.</p>
    <p>He laughed at that, and his laugh was merry and fr... in front.</p>
    <p>"Canary-birds?" he asked.</p>
    <p>"Oh no; it's just Eureka, my kitten. I thought that was the best way to
carry her."</p>
    <p>The boy nodded.</p>
    <p>"Eureka's a funny name for a cat," he remarked.</p>
    <p>"I named my kitten that because I found it," she explained. "Uncle Henry
says 'Eureka' means 'I have found it.'"</p>
    <p>"All right; hop in."</p>
    <p>She climbed into the buggy and he followed her. Then... , and said "Gid-
dap!"</p>
    <p>The horse did not stir. Dorothy thought he just wiggled one of his
drooping ears, but that was all.</p>
    <p>"Gid-dap!" called the boy, again.</p>
    <p>The horse stood still.</p>
    <p>"Perhaps," said Dorothy, "if you untied him, he would go."</p>
    <p>The boy laughed cheerfully and jumped out.</p>
  </Spoonful>
  <Spoonful id="4">
    <p>"Guess I'm ...the long nose of the animal.</p>
    ...
last.</p>
  </Spoonful>
</SpoonreadBody>
```

Fig. 9

Q&A Structure:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Validation>
  <Chapter id="1">
    <Spoonful id="1">
      <Question id="1" type="MultipleChoice" answer="3">
        <q>What was tucked under the girl's arm?</q>
        <a id="1">a parrot</a>
        <a id="2">a pillow</a>
        <a id="3">a parasol</a>
        <a id="4">a teddy bear</a>
        <renderToDom>true</renderToDom>
      </Question>
    </Spoonful>
    <Spoonful id="2">
      <Question id="1" type="MultipleChoice" answer="1">
        <q>The_____seemed almost new, for it had a shiny top and side
curtains.</q>
        <a id="1">buggy</a>
        <a id="2">cart</a>
        <a id="3">wagon</a>
        <a id="4">Prius</a>
        <renderToDom>true</renderToDom>
      </Question>
    </Spoonful>
    <Spoonful id="3">
      <Question id="1" type="MultipleChoice" answer="2">
        <q>What does Dorothy carry in her bird cage?</q>
        <a id="1">parakeet</a>
        <a id="2">kitten</a>
        <a id="3">puppy</a>
        <a id="4">hamster</a>
      </Question>
    </Spoonful>
    ...
  </Chapter>
</Validation>

```

Fig. 10

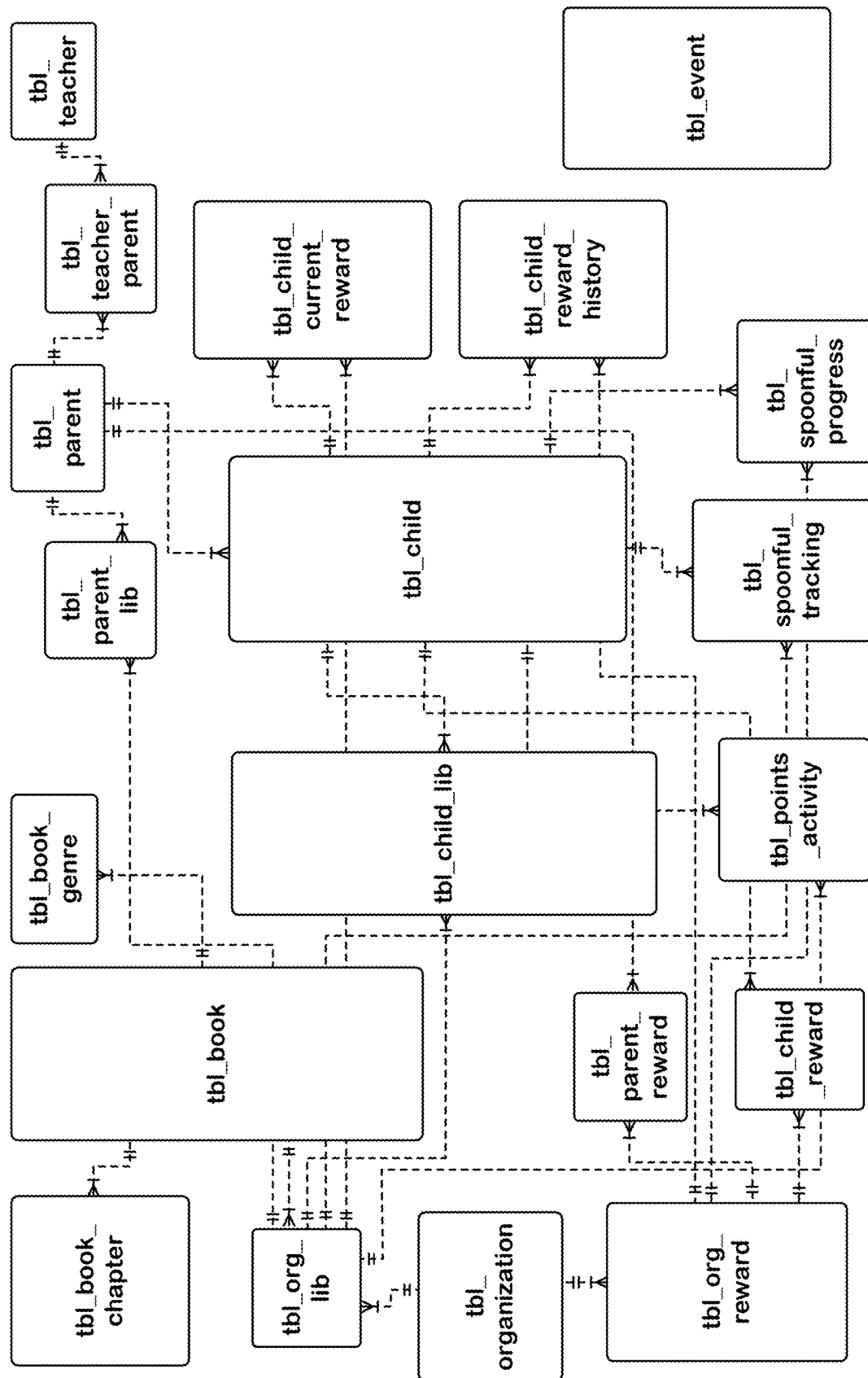


Fig. 11

```
CREATE TABLE tbl_book (  
    book_id  
    title  
    sub_title  
    author_id  
    digital_id  
    short_desc  
    long_desc  
    edition  
    pub_date  
    added_date  
    book_level  
    ref_book_status_id  
    ref_book_purchase_type_id  
    price  
    rating  
    cover_image_url  
    content_file_name  
    validation_metadata  
    chapter_cnt  
    spoonful_cnt  
    word_cnt  
    question_cnt  
    ref_publisher_id  
    is_content_final  
  
    ...  
);
```

Fig. 12

```
CREATE TABLE tbl_book_chapter (  
    book_chapter_id  
    book_id  
    chapter_id  
    chapter_file_name  
    chapter_content  
    validation_metadata  
    spoonful_count  
    word_count_info  
    ...  
);
```

Fig. 13

```
CREATE TABLE tbl_child_current_reward (  
    current_reward_id  
    child_id  
    org_reward_id  
    ref_reward_status_id  
    requested_date  
    approved_date  
    claimed_date  
    fulfilled_date  
    denied_date  
    points_needed  
    points_redeemed  
    ...  
);
```

Fig. 14


```
CREATE TABLE tbl_child (  
  child_id  
  user_id  
  parent_id  
  age  
  grade_level  
  ref_reading_level_id  
  ref_spelling_performance_level_id  
  ref_comprehension_performance_level_id  
  ref_vocabulary_performance_level_id  
  favorite_movie  
  favorite_book1  
  favorite_book2  
  favorite_book3  
  favorite_genres  
  verification_code  
  code_exp  
  points_earned  
  points_redeemed  
  points_available  
  crystals_counter  
  ...  
);
```

Fig. 15

```
CREATE TABLE tbl_child_lib (  
    child_lib_id  
    child_id  
    org_lib_id  
    due_date  
    unit            ???  
    num_units  
    freq_in_days  
    delivery_interval_in_mins  
    delivery_start_time  
    delivery_end_time  
    assignment_date  
    reading_start_date  
    read_time  
    read_percentage  
    score  
    chapter_counter  
    spoonful_counter  
    word_counter  
    question_counter  
    accuracy_index  
    has_completed  
    completion_date  
    ...  
);
```

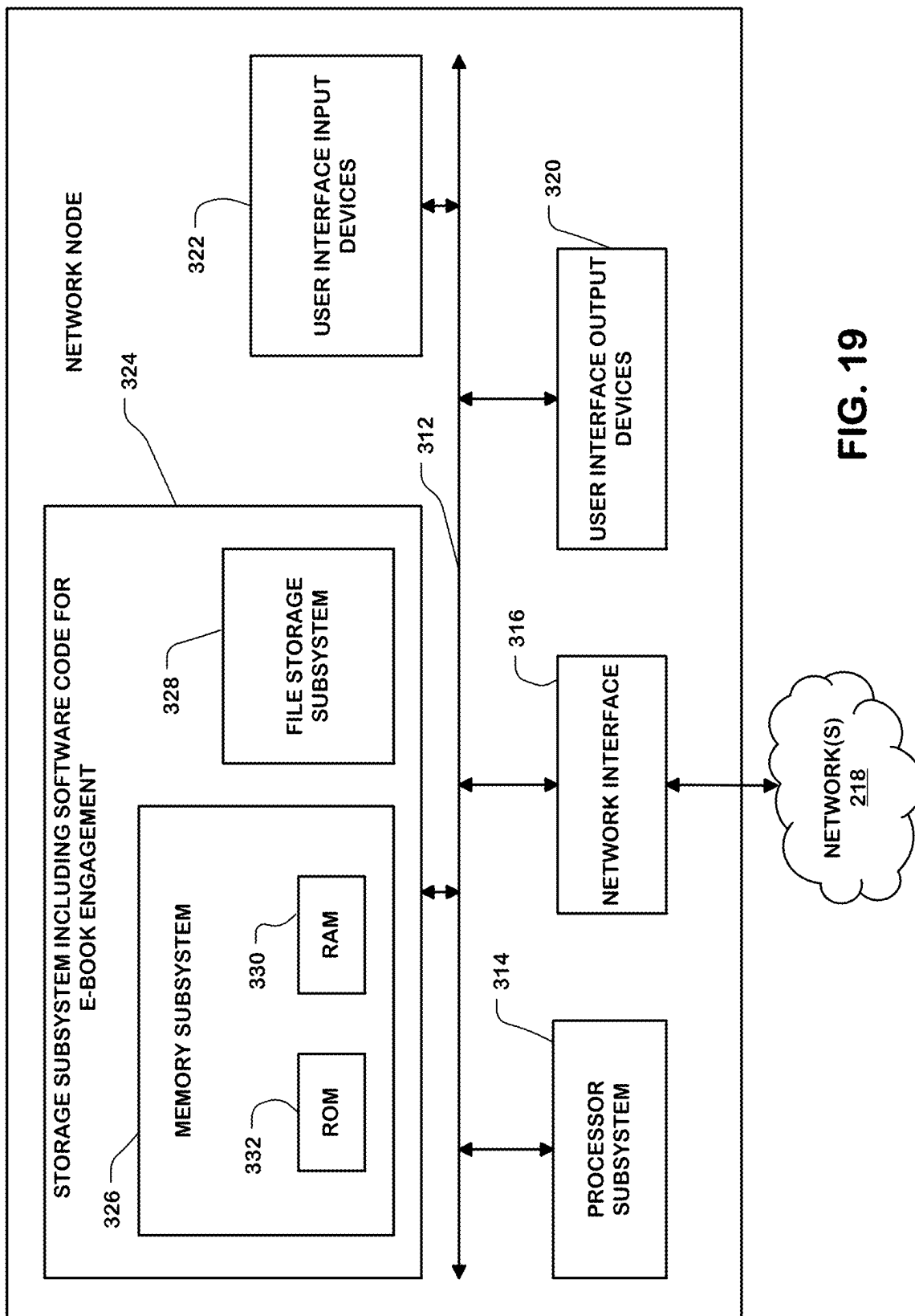
Fig. 16

```
CREATE TABLE tbl_spoonful_tracking (  
    spoonful_tracking_id  
    tracking_date  
    child_id  
    org_lib_id  
    chapter_id  
    spoonful_id  
    question_id  
    total_attempts  
    successful_attempts  
    read_time  
    points_value  
    ...  
);
```

Fig. 17

```
CREATE TABLE tbl_spoonful_progress (  
    spoonful_progress_id  
    progress_date  
    child_id  
    org_lib_id  
    num_units_completed  
    ...  
);
```

Fig. 18



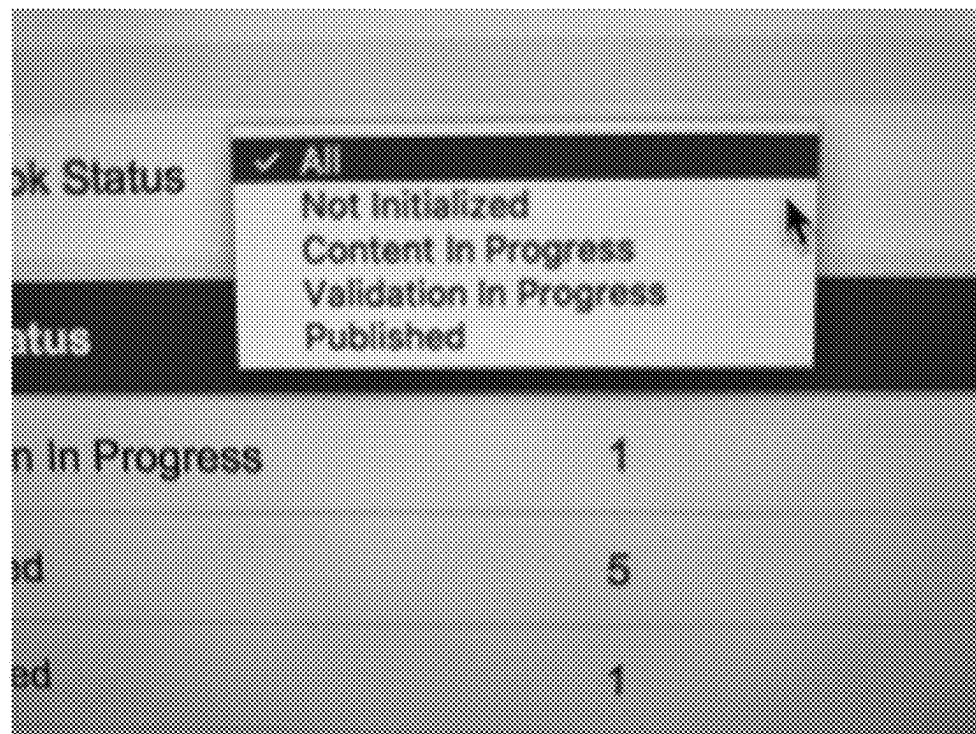
SPOONREAD

Home - Book List

Book List Book Status: All ▼ Add Book

#	Book Name	Author	Book Status	Chapters	Spoonfuls
1	Oedipus the King	Richard Jebb	Validation In Progress	1	84
2	The Tragedy of Julius Caesar	William Shakespeare	Published	5	177
3	Appley Dappley's Nursery Rhymes	Beatrix Potter	Published	1	14
4	Cecily Pansley's Nursery Rhymes	Beatrix Potter	Published	1	14
5	The Story of Miss Muppet	Beatrix Potter	Published	1	14
6	The Tale of Johnny Town-Mouse	Beatrix Potter	Published	1	26
7	The Tale of Tommy Tipsoo	Beatrix Potter	Published	1	26
8	The Tale of Mrs. Tittlemouse	Beatrix Potter	Published	1	26
9	The Tale of Jemima Puddle- Duck	Beatrix Potter	Published	1	26
10	The Tale of Ginger and Pickles	Beatrix Potter	Published	1	26
11	The Story of A Fierce Bad Rabbit	Beatrix Potter	Published	1	14
12	The Tale of The Flopsy Bunnies	Beatrix Potter	Published	1	26
13	The Tale of Tom Kitten	Beatrix Potter	Published	1	26
14	The Tale of Mr. Jeremy Fisher	Beatrix Potter	Published	1	26
15	The Pie and the Patty-Pan	Beatrix Potter	Published	1	23
16	The Tale of Mrs. Tiggy-Winkle	Beatrix Potter	Published	1	26
17	The Tale of Two Bad Mice	Beatrix Potter	Published	1	26
18	The Tailor of Gloucester	Beatrix Potter	Published	1	26
19	Robinson Crusoe	Daniel Defoe	Validation In Progress	20	508
20	Grimm's Household Tales, Volume 1	Jacob Grimm	Published	4	41
21	Cinderella or the Little Glass Slipper	George Datzel	Published	1	6
22	The Swiss Family Robinson	Johann Wyss	Validation In Progress	18	552
23	Rebecca of Sunnybrook Farm	Kate Wiggin	Content In Progress	31	374
24	White Fang	Jack London	Content In Progress	25	401
25	Just So Stories	Rudyard Kipling	Validation In Progress	12	175
26	Heidi	Johanna Spyri	Content In Progress	23	441
27	Black Beauty	Anna Sewall	Content In Progress	43	292
28	The Book of Dragons	Edith Nesbit	Validation In Progress	8	204
29	The Boy and the Badger	Edith Nesbit	Published	1	26

Fig. 20



- Not Initialized = not started
- Content in Progress = SpoonFul Generation
- Validation in progress = Questions and Answers
- Published = pushed out to book store

Fig. 21

SPOONREAD	
Home / Book List / Treasure Island	
Book Info	
Title	Treasure Island
Author Name	Robert Stevenson
SubTitle	
Status	Published
Short Description	Gold! Pirates! Treachery! Robert Louis Stevenson's thrilling tale of adventure on the high seas has it all. Join the young Jim Hawkins and his crew as they set sail in search of the hidden treasure of the infamous pirate Captain Flint. For sheer storytelling delight and pure adventure, Treasure Island has never been surpassed. From young Jim Hawkins's first encounter with the sinister beggar Pew to the climactic battle with the most memorable villain in literature, Long John Silver, this novel has fired readers' imaginations for generations. A rousing tale of treachery, greed, and daring, Treasure Island continues to enthrall readers of all ages.
Long Description	Gold! Pirates! Treachery! Robert Louis Stevenson's thrilling tale of adventure on the high seas has it all. Join the young Jim Hawkins and his crew as they set sail in search of the hidden treasure of the infamous pirate Captain Flint. For sheer storytelling delight and pure adventure, Treasure Island has never been surpassed. From young Jim Hawkins's first encounter with the sinister beggar Pew to the climactic battle with the most memorable villain in literature, Long John Silver, this novel has fired readers' imaginations for generations. A rousing tale of treachery, greed, and daring, Treasure Island continues to enthrall readers of all ages.
Edition	
Digital Id	
Book Level	8.3
Book Purchase Type	Free
Price	0
Rating	3.5
Cover Image Url	http://spoon-read.com/3000/images/cover_image/7/TreasureIsland.jpg
Chapter Count	34
Word Count	87527
Question Count	374
Paragraph Count	374
Content file	Treasure_Island.zip
Validation xml file	xml data
Full Information Download Full Book Download PDF	

Fig. 22

SPOONREAD

Home / Book List / Treasure Island

Edit Book Info

Title	Treasure Island
Author Name	Robert Stevenson
SubTitle	Enter SubTitle
Status	Published
Short Description	Gold! Pirates! Treachery! Robert Louis Stevenson's thrilling tale of adventure on the high seas has it all. Join the young Jim Hawk
Long Description	Gold! Pirates! Treachery! Robert Louis Stevenson's thrilling tale of adventure on the high seas has it all. Join the young Jim Hawk
Edition	Enter Edition
Digital Id	Enter Digital Id
Book Level	8.3
Book Purchase Type	Free
Price	\$
Rating	3.5
Cover Image Url	http://spoon-read.com/3000/images/cover_image/7/TreasureIsland.jpg
Chapter Count	34
Word Count	67527
Question Count	374
Paragraph Count	374
Content file upload	Browse... No file selected.
Validation Metadata	

Fig. 23

SPOONREAD

Home Books Add Book

Add Book

Title	Title	
Sub Title	Sub Title	
Short Description	Short Description	
Long Description	Long Description	
Edition	Edition	
Book Level	Book Level	
Image	Browse...	No file selected.
Chapter Count	Chapter Count	0
Paragraph Count	Paragraph Count	0
Word Count	Word Count	0
Question Count	Question Count	0
Author	Select Author	
Book Genre	Select Genre	
Book Status	Not initialized	
Purchase Type	Free	
Price	Price	
Rating	Rating	

Fig. 24

SPoonREAD

Home Book List Treasures Island Book Chapter List

Book Info

#	Chapter File Name
1	Chapter01
2	Chapter02
3	Chapter03
4	Chapter04
5	Chapter05
6	Chapter06
7	Chapter07
8	Chapter08
9	Chapter09
10	Chapter10
11	Chapter11
12	Chapter12
13	Chapter13
14	Chapter14
15	Chapter15
16	Chapter16
17	Chapter17
18	Chapter18
19	Chapter19
20	Chapter20
21	Chapter21
22	Chapter22
23	Chapter23
24	Chapter24
25	Chapter25
26	Chapter26
27	Chapter27
28	Chapter28
29	Chapter29

Fig. 25

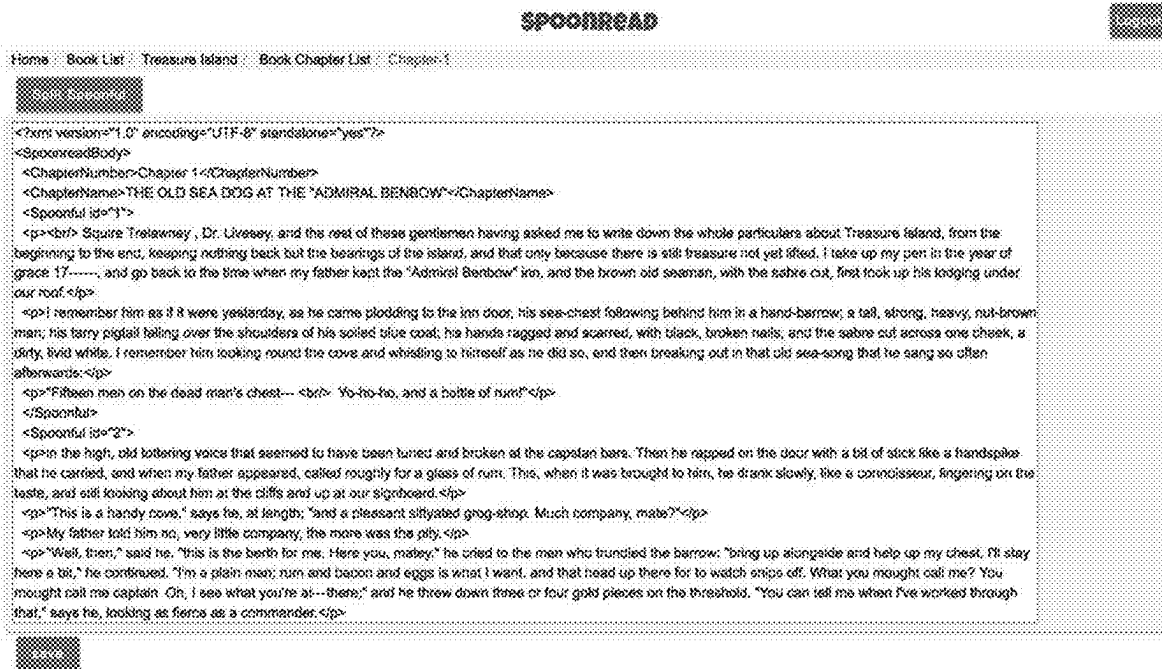


Fig. 26

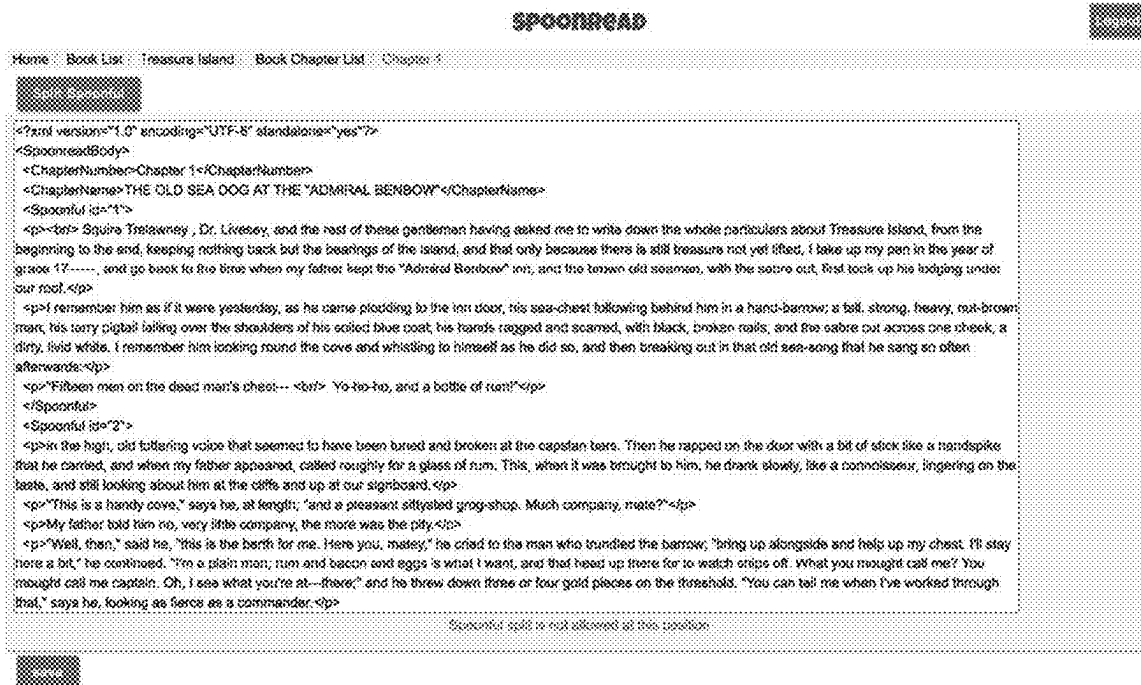


Fig. 27

SPOONHEAD

HomeBook ListTreasure IslandValidation

Chapters

#	Chapter Name	Spoonful Count
1	Chapter01	12
2	Chapter02	11
3	Chapter03	12
4	Chapter04	12
5	Chapter05	10
6	Chapter06	12
7	Chapter07	10
8	Chapter08	10
9	Chapter09	9
10	Chapter10	11
11	Chapter11	11
12	Chapter12	10
13	Chapter13	9
14	Chapter14	11
15	Chapter15	12
16	Chapter16	9
17	Chapter17	8
18	Chapter18	8
19	Chapter19	11
20	Chapter20	10
21	Chapter21	11
22	Chapter22	11
23	Chapter23	9
24	Chapter24	11
25	Chapter25	8
26	Chapter26	18
27	Chapter27	12
28	Chapter28	16
29	Chapter29	12

Fig. 28

SPOONREAD

Home Book List Treasure Island Validation Chapter 1

Chapter info

Spoonful 1	1
Spoonful 2	1
Spoonful 3	1
Spoonful 4	1
Spoonful 5	1
Spoonful 6	1
Spoonful 7	1
Spoonful 8	1
Spoonful 9	1
Spoonful 10	1
Spoonful 11	1
Spoonful 12	1

Fig. 29

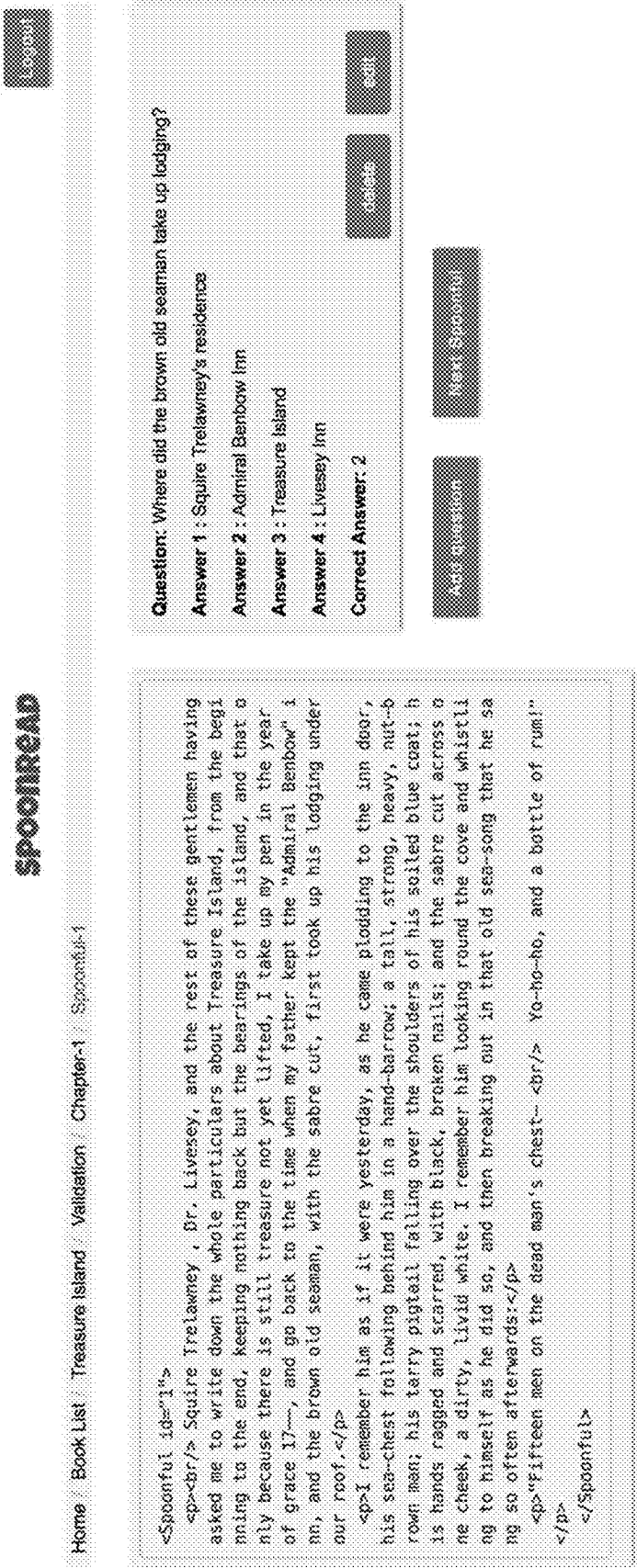


Fig. 30

Home

Book List

Treasure Island

Validation

Chapter 1

Spoonful 1

Spoonful 1

<Spoonful id="1">
<p>
 Squire Trelawney , Dr. Livesey, and the rest of these gentlemen having asked me to write down the whole particulars about Treasure Island, from the beginning to the end, keeping nothing back but the bearings of the island, and that only because there is still treasure not yet lifted, I take up my pen in the year of grace 17—, and go back to the time when my father kept the "Admiral Benbow" inn, and the brown old seaman, with the sabre cut, first took up his lodging under our roof.</p>
<p>I remember him as if it were yesterday, as he came plodding to the inn door, his sea-chest following behind him in a hand-barrow; a tall, strong, heavy, nut-brown man; his tarry pigtail falling over the shoulders of his soiled blue coat; his hands ragged and scarred, with black, broken nails; and the sabre cut across one cheek, a dirty, livid white. I remember him looking round the cove and whistling to himself as he did so, and then breaking out in that old sea-song that he sang so often afterwards:</p>
<p>"Fifteen men on the dead man's chest—
 Yo-ho-ho, and a bottle of rum!"</p>
</Spoonful>

Logout

Question: Where did the brown old seaman take up lodging?

Answer 1: Squire Trelawney's residence

Answer 2: Admiral Benbow Inn

Answer 3: Treasure Island

Answer 4: Livesey Inn

Answer 5: sample text

Answer 6: sample text

Answer 7: sample text

Answer 8: sample text

Answer 9: sample text

Answer 10: sample text

Correct Answer: 2

Cancel

Save

Navigation

Next Question

Fig. 31

**ELECTRONIC DOCUMENT PRESENTATION
MANAGEMENT SYSTEM****PRIORITY APPLICATIONS**

This application is a continuation of U.S. Non-Provisional patent application Ser. No. 18/807,883 titled "Electronic Document Presentation Management System," filed Aug. 16, 2024 which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/671,509, filed Feb. 14, 2022, now U.S. Pat. No. 12,067,900, issued Aug. 20, 2024 which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/037,523, filed Sep. 29, 2020, now U.S. Pat. No. 11,250,718, issued Feb. 15, 2022 which is a continuation of U.S. Non-Provisional patent application Ser. No. 15/798,248, filed Oct. 30, 2017, now U.S. Pat. No. 10,796,591, issued Oct. 6, 2020 which claims the benefit of U.S. Provisional Patent Application No. 62/569,285, titled "Electronic Document Presentation Management System," filed Oct. 6, 2017 and U.S. Provisional Patent Application No. 62/484,275, titled "Spoonread Systems And Methods," filed Apr. 11, 2017. The Non-Provisional and Provisional applications are incorporated herein by reference.

This application is also a continuation of U.S. Non-Provisional patent application Ser. No. 18/807,891 titled "Electronic Document Presentation Management System," filed Aug. 16, 2024 which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/671,505, filed Feb. 14, 2022, now U.S. Pat. No. 12,067,899, issued Aug. 20, 2024 which is a continuation of U.S. Non-Provisional patent application Ser. No. 16/685,685, filed Nov. 15, 2019, now U.S. Pat. No. 11,250,717, issued Feb. 15, 2022 which is a continuation of U.S. Non-Provisional patent application Ser. No. 15/798,248, filed Oct. 30, 2017, now U.S. Pat. No. 10,796,591, issued Oct. 6, 2020 which claims the benefit of U.S. Provisional Patent Application No. 62/569,285, titled "Electronic Document Presentation Management System," filed Oct. 6, 2017 and U.S. Provisional Patent Application No. 62/484,275, titled "Spoonread Systems And Methods," filed Apr. 11, 2017. The Non-Provisional and Provisional applications are incorporated herein by reference.

**REFERENCE TO COMPUTER PROGRAM
LISTING APPENDIX**

A computer program listing appendix submitted electronically via EFS-Web in ASCII text accompanies this application and is incorporated by reference. The name of the ASCII text file is "SPRD_Computer_Program_Appx," created on Oct. 25, 2017 and is 12,416 bytes.

BACKGROUND**Technological Field**

The invention generally relates to the field of electronic books (e-books) and in particular to a framework for tools for interacting with e-books and other electronic documents.

Description of Related Art

E-books have become an important channel for distribution of publications, including fiction and non-fiction literature of virtually all types. The technology is adaptable to enable linking of many types of enrichment tools to content of the publications, including dictionaries, internet searching tools, text searching tools, advertising channels, and so on.

See for example U.S. Pat. No. 9,678,634, titled "Extensible Framework For Ereader Tools," by Murray et al., issued 13 Jun. 2017.

One limitation on growth of e-book publishing arises with respect to use in an educational context, including use by children. Also, other than some special purpose e-readers, the platforms used for reading the e-books are often mobile phones and tablets that are not well adapted to the display of content from e-books. It is desirable for example to display segments of the e-books in a way that preserves literary integrity expressed by the author in paragraph and chapter boundaries of the text. However, the limited display available on a portable phone, for example, can present a technological limit on the segment displayed in terms of having a large enough font for the reader and a portion of the text that can express a substantial fact or concept in the work. The widely variant sizes of paragraphs and chapters as formed by the authors makes these literary boundaries unsuited in a technological sense to be utilized for defining the segment of the work being displayed on these kinds of platforms. This incongruity contributes to discomfort and inefficiency in the reading experience, and deters use of small format displays.

It is desirable to provide a technology that can extend the use of e-books for children, in educational contexts and on mobile devices.

SUMMARY

A technology is described for managing engagement and presentation of an electronic document such as an e-book in a way that makes it both efficient on small platform devices, and engaging in a way that can encourage use of the devices. The technology includes an electronic document including metadata that partitions the electronic document for engagement using a reader application, and a computer system architecture solving problems related to validating traversal of electronic documents in computer systems, and supporting utilization of the electronic document in a game-like process.

A computer executed reader application is described that utilizes the electronic document and metadata, and manages presentation of the content of the electronic document. The reader application in an embodiment described herein executes a process that includes retrieving an electronic document including content and container metadata. The container metadata identifies engagement containers within the electronic document that comprise respective segments of the content. The container metadata is metadata that is distinct from or in addition to metadata that identifies literary boundaries, such as paragraph and chapter boundaries. The procedure also includes retrieving engagement data linked to the engagement containers in the electronic document. The engagement data comprises parameters of an engagement based on the segments of content in the linked engagement containers. The engagement defined by the engagement data comprises a user interface sequence by which a reader validates review of the segment of content, such as a challenge/response (e.g., question and answer) sequence, based on content of the linked engagement container. The reader application includes a procedure to traverse engagement containers identified in the container metadata of a particular electronic document. The procedure includes for a current engagement container, displaying content of the current engagement container on the display, and accepting from user input at the platform, an engagement signal for initiation of engagement of the current

engagement container, such as by clicking a button on a touchscreen interface to indicate that the reader has reviewed the content. The procedure includes, after receipt of the engagement signal for the current engagement container, enabling an engagement tool on the platform. The engagement tool presents prompts on the display for the engagement based on the parameters defined in the engagement data linked to the current engagement container. The procedure then accepts user input at the platform in response to the prompts presented in the engagement. After receipt of the data in response to the prompts, the procedure selects a next engagement container in the electronic document, and can assign points for the current engagement container based on the received data and the parameters defined in the engagement data. Also, the procedure sends a notification to the server-side node validating traversal of the current engagement container.

The reader application can include a pacing procedure used to enable display of a selected engagement container. Thus, in selection of a next engagement container, the pacing procedure is applied. The pacing procedure retrieves parameters defining an assigned pace for completing the particular electronic document, such as a target assignment completion date or a specified number of chapters or paragraphs in the document. Also, the pacing procedure can apply parameters defining a time of day during which access to the particular electronic document is allowed. The pacing procedure can enable or disable presentation of a selected engagement container based on the pacing parameters. Thus, if the selection of the current engagement container occurs outside the allowed time of day, then the reader procedure can present a notice to the reader indicating that access is prevented during this time. Also, if the reader procedure has traversed a number of engagement containers that correspond with an assigned number of chapters or paragraphs to be read in a given day or other time period, then the reader procedure can present a notice to the reader indicating that access is prevented because of the pacing limitations.

A computer system architecture described herein comprises a server application executed using one or more server-side nodes, and two types of applications executed on client-side nodes. One of the types of client-side applications is characterized as a supervisor application controlled for example by a parent, teacher or tutor, and the other type of client-side applications is characterized as a reader application utilized by a student or child to read the electronic document.

The server-side node executes a server application including computer programs and database tools for executing a server process. The server process includes maintaining access to a library of electronic documents, such as e-books, that include content and container metadata in addition to markup like paragraph and chapter boundaries. The container metadata identifies engagement containers within the electronic document that comprise respective segments of the content. This process also includes maintaining access to engagement data for the electronic documents in the library. The engagement data is linked to the engagement containers in the electronic documents and comprises parameters of an engagement based on the segment of content in the linked engagement containers. Engagement defined by the engagement data can comprise a question and a response, where the system at the reader presents the question as a prompt or challenge in the display, and the user inputs the answer as a response to the challenge, to validate that a reader had reviewed the segment of content defined by the linked engagement container.

The process executed at the server can include maintaining supervisor account records. The supervisor account records identify a set of reader accounts under supervision. The process executed at the server can also include maintaining reader account records. The reader account records can store identifiers of electronic documents assigned to the reader accounts, and indications of progress in the assigned electronic documents that is determined by monitoring the engagement on each engagement container.

The server application can include executing an application program interface including parameters and procedures executed using the server-side node by communication with a network node or network nodes executing supervisor applications and reader applications.

The API can include a procedure or procedures in communication with the supervisor applications that add reader accounts to the set of reader accounts in the supervisor account record, to add particular electronic documents to the reader accounts in the set of reader accounts, to add parameters of assignments of the particular electronic documents to reader accounts and provide access to a particular supervisor application to reader account records of reader accounts added by the particular supervisor application.

The API can include a procedure or procedures in communication with the reader applications to provide access to the particular electronic documents in the reader accounts, to provide access to the engagement data for the particular electronic documents in reader accounts, to deliver parameters of the assignments of particular electronic documents in the reader accounts, to deliver parameters indicating progress in the assigned electronic documents, and to receive notifications from the reader applications of progress in the particular assigned electronic documents.

According to a configuration described herein, the system includes a supervisor application having a unique login, and a reader application configured for use by a child or student without sharing personal identifying information with the presentation system in general. The system is configured for traversing the electronic document in a manner that enables validation of progress and promotes engagement by the reader, and a server which maintains data sets used for delivering electronic documents, tracking progress in the electronic documents by the readers, delivering status information to the supervisor application and to the reader application.

These tools are configured to manage presentation of electronic documents in the manner of running computer implemented games, where the games involve traversing electronic documents, including presenting content to the readers in "spoonfuls," performing engagement exercises linked to the spoonfuls that validate reading of the electronic documents, and executing algorithms for awarding points based on the quality and success of the performance of engagement exercises.

The supervisor application in an embodiment described herein includes tools for assigning electronic documents to specific reader apps under its supervision. Also, the supervisor application enables the establishment of performance parameters, such as the amount of time within which completion of the reading of the electronic document must be completed, time of day windows in which the reader application can be engaged, and other performance parameters. The supervisor application also enables assigning rewards to readers that encourage engagement with the assigned electronic documents.

The procedure at the reader application includes notifying the server of progress through the electronic document,

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including the engagement containers reviewed, the score assigned to each engagement container, the time duration in which the engagement container was displayed, and other parameters of the engagement.

Embodiments of the technology include maintaining an accumulated point total for the electronic document by combining the points awarded upon validation of the engagement containers traversed. The points can have positive or negative values depending on parameters specified for the engagement by the engagement metadata. There can be a plurality of engagement types presented by the engagement tool, where the engagement data can assign an engagement type to linked engagement containers.

In support of the presentation system, a delivery mechanism is provided herein that is embedded inside the electronic document using metadata tags to define engagement containers which enclose a “spoonful” of content. The system includes data structures binding the engagement containers to engagement data such as questions and answers and other components associated with the spoonful that support the presentation. Images, paragraph, chapter and other style features designed by the author and related to the literary integrity of the electronic document can be left unaltered in the electronic document overlaid by the metadata described herein that supports the engagement process during presentation.

In an embodiment described, different access authorization procedures are used for access to the reader account record and access to the supervisor account record.

In a system described herein, an application program interface API comprising a set of parameters and procedures is implemented by a server with complementary procedures in the supervisor application and the reader app, enabling the server to communicate with a supervisor application and a reader application to support presentation of the electronic document, to maintain data about status of electronic documents, and to support maintenance and delivery of engagement parameters. In this manner, progress in the electronic documents is maintained apart from the reader app, enabling portability of the engagement with the electronic document from platform to platform without loss of status information, and enabling real-time and off-time monitoring of progress using the supervisor application.

Implementation of the technology described herein includes storage of computer programs in memory, based on a non-transitory data storage medium or data storage media, the computer programs configured for execution by data processors to perform the procedures of one or more of the supervisor app, the reader application and the server. In some embodiments, the memory is part of a network node configured to execute one or more of the procedures described herein. In other embodiments, the memory is a portable memory device.

Other aspects and advantages of the technology described herein can be seen on review of the drawings, the detailed description and the claims, which follow.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a network perspective diagram illustrating configuration of a system for managing presentation of electronic documents based on engagement containers as described herein.

FIG. 2 is a network perspective diagram illustrating a configuration of the system for intake and processing of electronic documents, such as e-books, to provide metadata

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identifying engagement containers and engagement data for use in the engagement system of FIG. 1.

FIG. 3 is a simplified diagram of procedures of a reader application in an embodiment described herein.

FIG. 4 is a simplified diagram of procedures in a supervisor application in an embodiment described herein.

FIG. 5 illustrates a set of user interface screens utilized by the reader application in the procedure for presentation of electronic documents as an example for use in a system as described herein.

FIG. 6 illustrates a set of user interface screens utilized by the supervisor application in the procedure for presentation of an electronic document as an example for use in a system as described herein.

FIG. 7A is a flowchart of parts of a procedure executed by a reader application as described herein for traversing an electronic document.

FIG. 7B is a flowchart of parts of a procedure executed by a reader application as described herein for traversing an electronic document.

FIG. 8A is a flowchart of parts of a procedure executed by a supervisor application as described herein, for assigning electronic documents to readers under its supervision.

FIG. 8B is a flowchart of other parts of a procedure executed by a supervisor application as described herein, for assigning electronic documents to readers under its supervision.

FIG. 9 illustrates an example portion of an electronic document, with engagement container metadata.

FIG. 10 illustrates an example of engagement data linked to engagement containers in an electronic document, such as that of FIG. 9.

FIG. 11 constitutes an entity relationship diagram for data structures maintained in support of the procedure for presentation of electronic documents as described herein.

FIGS. 12, 13, 14, 15, 16, 17 and 18 illustrate contents of data structures in FIG. 11, for one example implementation.

FIG. 19 is a simplified diagram of a hardware platform which is configured for execution of the server or other apps involved in the e-book engagement system described herein.

FIGS. 20-31 are screen shot images of a user interface for an editor application used in support of assigning container metadata and engagement data for electronic documents as described herein, and in which:

FIG. 20 illustrates display of a book list based on data retrieved using the API from the server indicating information about the electronic documents, including book status.

FIG. 21 illustrates a pulldown menu for optional book status entries using the editor application, including a “not initialized” status for electronic documents for which the editing process has not started, a “content in progress” status for electronic documents for which the generation of the container metadata identifying spoonfuls is in progress, a “validation in progress” status for which the generation of validation data, such as questions and answers, linked to the engagement containers is in progress, and a “published” status indicating that the electronic document is ready for use in the presentation system.

FIG. 22 illustrates content derived from the server for information about an e-book which can be edited using the editor app.

FIG. 23 shows a user interface enabling editing of the book information.

FIG. 24 shows a user interface supporting entry of additional books into the system.

FIG. 25 illustrates menus for selecting or editing information about chapters in e-books.

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FIG. 26 is a screenshot supporting the adjustment of, or addition of, engagement container metadata to an electronic document by the editor app.

FIG. 27 is like FIG. 26, showing that the editor app can enforce editing rules that define locations in which splitting of an engagement container are allowed or not allowed.

FIG. 28 illustrates menus for selecting chapters and spoonfuls for the purposes of adding or editing validation data.

FIG. 29 illustrates menus for selecting spoonfuls after selection using the menu of FIG. 28, for the purposes of adding or editing engagement data.

FIG. 30 illustrates a display of content of an engagement container along with engagement data with tools prompting editing of the engagement data.

FIG. 31 shows display of a user input screen configured for entry of engagement data based on questions and multiple choice answers which can be linked to corresponding engagement containers.

DETAILED DESCRIPTION

A detailed description of embodiments of the technology is provided with reference to the FIGS. 1-31.

A flexible technology platform that chunks content and generates an engagement to validate and confirm comprehension is described herein. The platform provides for frequent, embedded, real-time monitoring of understanding which can dramatically increase the impact and efficacy of student's later studying and final review. Also, the platform provides a combination of embedded comprehension checks with study and final review, which is highly effective for both fiction and non-fiction content.

The electronic document presentation system described herein provides a computer implemented device around reading, many of the layers of which come together to create the look of a game. Turning the book into the core of the game can be expressed in the following algorithm.

```
(MobileDeviceType+MobileDeviceMetaData+Book+
  BookMetaData=MethodOfValidation·      Variation-
  OfMethodOfValidation)
```

The combinatorial effect of executing the object "MethodOfValidation" creates engagement with the electronic document that encourages reading. "VariationOfMethodOfValidation" creates an infinite number of possibilities for validation of reading which constitutes game activity.

Game activity is on a per "spoonful," per chapter, and per book basis, where a spoonful is a segment of content identified by container metadata that is distinct from and in addition to metadata identifying boundaries set by an author that preserve literary integrity like paragraphs and chapters. The outcome of the validation process has direct influence on points gained or lost in the game, and transcends any single book. Therefore, moving through e-books, is like moving through levels of a game or rooms in an adventure game where each e-book is the room. Each e-book is its own discrete play activity but feeds into the larger total of points thereby making traversing the e-book the primary object of the game.

The validation process can be equivalent to bullets in a single person shooter genre game. An action by a player either gets validated (dodge the bullet) or fails the validation (get hit by the bullet) so both positive and negative emotions of a game are generated along with adding and subtracting of points for getting hit by or dodging a bullet.

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Then, it can be said that the game results are derived using the following formula:

SINGULAR_RESULT=MethodOfValidation. Variation-OfMethodOfValidation

Where,

SINGULAR_RESULT:

Pass=Full Points

Fail=Partial Points|No Points

So that

CUMULATIVE_RESULT=SINGULAR_RESULT++

[key: ++ means incremented based on itself]

Then, When

CUMULATIVE_RESULT>=PRIZE_BOUNTY

GAME_STATUS=WON

If CUMULATIVE_RESULT<PRIZE_BOUNTY

GAME_STATUS=IN PROGRESS

Additionally,

SINGULAR_RESULT<SINGULAR_RESULT_ PER-
FORMANCE_METRIC

LIFE_LEVEL=LIFE_LEVEL -1 (There can be an overall heartbeat that stays steady with good validation results and decreases with poor validation results, after losing health it is possible to gain health back with good reading)

SINGULAR_RESULT<SINGULAR_RESULT_ PER-
FORMANCE_METRIC

LIFE_LEVEL=LIFE_LEVEL+1

The LIFE_LEVEL in the game is also a hook back into the messaging part of the app, for example, if LIFE_LEVEL increases and decreases are trending certain ways, inspirational messages can be sent to the reader.

IF LIFE_LEVEL_DELTA_TRIGGER=TREND_
DESCRIPTOR

THEN SEND_INSPIRATIONAL_MESSAGE

FIG. 1 is a network perspective of a system for presentation of electronic documents as described herein. The network includes a network node configured to execute an engagement server 100 coupled via a network 118 such as the Internet with a plurality of network nodes configured to execute reader applications and supervisor applications. The network nodes executing the reader applications and the supervisor applications can comprise smart phones 120, tablets 130, laptop or personal computers 140, and other computer systems coupled to the network 118. The system is configured for presentation of the content using the reader application, assignment and tracking of the progress using the supervisor application, and management of the data and administration of the system using the server 100.

In FIG. 1, components of the server 100 are illustrated. In this example, the server includes an API 106 for coordinating utilization of a set of reader procedures 102 supporting the reader application, and a set of supervisor procedures 103 supporting the supervisor application. Also, the server 100 includes data sets 110, 111 in support of the system. The data sets in this example includes a data set 110 of electronic documents such as e-books and a data set 111 including metadata supporting the presentation of the electronic documents including container metadata, and engagement data and metadata as described herein. Also the data set includes files maintaining user accounts 112 for readers and supervisors, and files for managing notifications 113 among the elements of the system. Also, the system includes files identifying rewards made available to users of the applications. The server 100 in this example maintains an e-commerce link 115 which can be engaged for the purposes of acquiring electronic documents such as e-books from publishers, and acquiring or fulfilling the delivery of rewards earned in some embodiments by accumulating points authorized by the supervisor application.

FIG. 2 illustrates another aspect of the system by which electronic documents are acquired and configured for utilization. In this system, a publisher **161** of electronic documents is coupled to the network **118**. Engagement server **100** includes publication processing procedures **160** configured to process an electronic document to add container metadata and engagement data for the purposes of supporting the presentation of the electronic document as described herein. Also, the server **100** includes an API **106** configured for communication with an editor application which can execute on a network node **150** coupled with a network **118**. The publication processing procedures **160** produce metadata associated with the content of the electronic documents stored in the data set (e-books) **110** and (metadata) **111** utilized by the server **100**.

Standard metadata associated with electronic documents such as e-books is not effective for some electronic documents for defining engagement units for the purposes of the presentation system described herein. The amount of information in chapters and paragraphs varies widely and is completely under control of the authors. Thus, relying on these markers in the electronic documents for the purposes of engagement can be impractical. Thus, the electronic documents are further processed in technology described herein to identify engagement units referred to herein as spoonfuls, and tagged with metadata to define the engagement containers within the electronic document having an amount or type of content that satisfies requirements of engagement techniques applied. Addition of spoonful-based engagement container metadata to the electronic document makes computerized presentation more effective and efficient, and supports gamification of the presentation.

The spoonful generation is done in embodiments described herein on a “cleaned” version of each chapter file in the e-book, where markup associated with the e-book that may not be pertinent to the engagement system has been removed.

The procedures which can be executed at the server or other network nodes, used for defining the engagement containers can be adapted to the particular types of e-books. There are two algorithms described herein:

(A) For picture books, like the books in the Beatrix Potter collection.

A significant hallmark of these books is that they have many images, with very little text in between the images. The rules driving this algorithm are:

- A-1) Each spoonful can have a maximum of 2 images
- A-2) Each spoonful will start with an image, followed by the text until the next image (which will be placed in the next spoonful), OR it will start with text and end with an image.
- A-3) Corollary: since the amount of text in between the images in these books is small, each block of text will not be split across multiple spoonfuls.

(B) For books containing primarily text and a few embedded images.

The spoonful generation process takes a parameter: “data-size”. The default value is 1200 characters. This size has been empirically determined to effectively define an engagement unit with sufficient content for the purposes of a meaningful engagement with the reader. The size, however, typically does not match up with chapter and paragraph markup in the electronic document, and so further processing is necessary.

The rules for an example of this algorithm are:

B-1) The e-books generally have all the text enclosed within <p> (paragraph) tags at the top-level. Text within any single <p> tag will not split across spoonfuls.

B-2) The number of characters in each spoonful can be between 90% and 110% of the data-size value (e.g. 1200+/-120).

B-3) As the text inside each <p> tag is added to the current spoonful, the character count is incremented.

B-4) If the current count is <90% of data-size, check the next <p> tag.

If adding the text of the next <p> tag will put the count over 110% of data-size, then don't include it.

Else, add it to the current spoonful, and continue the iterative algorithm.

In addition to the automated markup process by which the engagement containers are specified, engagement data, which is referred to herein as validation metadata, must be produced and linked with, preferably, each engagement container. A successful execution of the engagement by the reader validates that the reader has read the content of the engagement container and is ready to progress to a next engagement unit.

The server **100** provides an interface allowing editing by a person for the final tagged file—i.e. an administrator or editor console. The tagged document identifying the initial placement of markup specifying the engagement containers is provided to the editor application. The user interface is presented allowing an operator of the editor application to adjust the boundaries of the engagement containers. Also, the user interface provides for entry of questions and multiple choice answers for the purposes of the engagement tool. Also, other types of engagement can be specified, including questions with text answers, questions requiring audio response, and other types of engagements that can validate review of the content of the corresponding linked engagement container.

After addition of the engagement data links to the engagement containers, and potentially editing of the boundaries of the engagement containers, using the editor application, the electronic document is ready for utilization in the presentation system described herein.

FIGS. 20-31 herein illustrate aspects of the editor application for an embodiment of the presentation system.

Functions of the server **100**, the reader application and the supervisor application can be understood with reference to the following specification of an API suitable for use as API **106** in FIG. 1, defining parameters and procedures of the system in support of presentation of the electronic document.

The following table provides the definition of data elements referred to in the specification of the API below. This information can be understood further with reference to the entity relationship diagram shown in FIG. 11.

parentData	parentId, authenticatedUserID, firstName, lastName, nickName, hasAvatar, avatarName, hasImage, imagePath, imageFileName, childCount
childData	childId, parentId, authenticatedUserID, firstName, lastName, nickName, hasAvatar, avatarName, hasImage, imagePath, imageFileName, age, gradeLevel, readingLevel, spellingPerformanceLevel, comprehensionPerformanceLevel, vocabularyPerformanceLevel, favoriteMovie, favoriteBook1, favoriteBook2, favoriteBook3, favoriteGenres, pointsEarned, pointsRedeemed, pointsAvailable, crystalsCounter

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-continued

bookData	bookId, title, subTitle, authorName, shortDesc, longDesc, edition, pubDate, addedDate, bookGenre, bookLevel, price, rating, coverImageUrl, chapterCount, spoonfulCount, wordCount, questionCount, bookContentURI	5
parentBookData	bookId, title, subTitle, authorName, shortDesc, longDesc, edition, pubDate, addedDate, lib_addedDate, bookLevel, price, rating, coverImageUrl, chapterCount, spoonfulCount, wordCount, questionCount, bookContentURI	10
childBookData	bookId, title, subTitle, authorName, shortDesc, longDesc, edition, pubDate, addedDate, bookLevel, price, rating, coverImageUrl, chapterCount, spoonfulCount, wordCount, questionCount, bookContentURI	15
assignmentData	dueDate, assignmentDate, readingStartDate, unit, numUnits, freqInDays, deliveryInterval, deliveryStartTime, deliveryEndTime, readTime, readPercentage, score, chapterCounter, spoonfulCounter, wordCounter, questionCounter, accuracyIndex, hasCompleted, completionDate	20
teacherData	teacherId, userId, firstName, lastName, nickName, hasAvatar, avatarName, hasImage, imageFileName	25
rewardData	rewardId, rewardName, rewardCategory, shortDesc, longDesc, imageUrl, widgetUrl, ecommLink, specialInstructions, purchasePrice	30
childRewardData	childId, rewardId, rewardStatus, requestedDate, approvedDate, claimedDate, fulfilledDate, deniedDate, pointsNeeded, pointsRedeemed	35
rewardHistoryData	rewardHistoryId, childId, rewardId, rewardStatus, requestedDate, approvedDate, claimedDate, fulfilledDate, pointsRedeemed	40
eventData	eventId, fromId, fromUserType, eventType, message, eventDate, isNotificationRequired, notificationSentDate, eventStatus, objectTypeOne, objectIdOne, objectTypeTwo, objectIdTwo, objectTypeThree, objectIdThree	45
parentRewardData	parentId, rewardId, rewardName, rewardCategory, shortDesc, longDesc, imageUrl, widgetUrl, ecommLink, specialInstructions, purchasePrice	50

The following table defines reference values for certain reference elements referred to in the specification of the API provided below.

Reference Element	Reference Values	
userType	Parent, Child, Administrator, Teacher	
objectType	Book, Reward	
eventType	Book Assigned, Book Completed, Spoonful Arrived, Spoonful Complete, Spoonful Read, Reward Requested, Reward Approved, Reward Claimed, Reward FulfillStarted, Reward FulfillCompleted, Reward Denied	45
readingLevel	NotInitialized, Preprimer, Primer, First, Second, Third, Fourth, Fifth, Sixth, Seventh, Eighth, Ninth, Tenth, Eleventh, Twelfth	50
performanceLevel	NotInitialized, Beg, Mid, End	
deviceType	iOS, Android	
hasImage	true/false (or) 1/0	
hasAvatar	true/false (or) 1/0	
hasCompleted	true/false (or) 1/0	55

Elements of an example of an API, including procedures and parameters, are defined as follows, used for communication with the server by the supervisor application and with the reader application.

1. addParent (POST)

Header {orgUID}

Payload {userName(M), password(M), firstName(M), lastName(M)}

Response {success, parentId, authenticatedUserUID}

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2. sendEmailVerificationCode (POST)

Header {orgUID}

Payload {userName(M)}

Response {success}

3. validateEmailVerificationCode (POST)

Header {orgUID}

Payload {userName(M), verificationCode(M)}

Response {success}

4. updateParent (POST)

Header {orgUID, sessionToken}

Payload {parentId(M), nickName(M), hasAvatar(M), avatarName(M), hasImage(M), imageFileName(M)}

Response {success}

5. addChildToParent (POST)

Header {orgUID, sessionToken}

Payload {parentId(M), name(M), age(M), gradeLevel(M), readingLevel(M), spellingPerformanceLevel(M), comprehensionPerformanceLevel(M), vocabularyPerformanceLevel(M), crystalsCounter(M)}

Response {success, childId, authenticatedUserUID, verificationCode}

6. updateChild (POST)/*favoriteGenres is an array of genres id's, Ex:-[1,2,3]*/

Header {orgUID, sessionToken}

Payload {childId(M), name(O), age(O), gradeLevel(O), readingLevel(O), spellingPerformanceLevel(O), hasImage(O), imageFileName(O), spellingPerformanceLevel(O), comprehensionPerformanceLevel(O), vocabularyPerformanceLevel(O), favoriteMovie(O), favoriteBook1(O), favoriteBook2(O), favoriteBook3(O), favoriteGenres(O), crystalsCounter(O)}

Response {success}

7. loginParent (POST)

Header {orgUID}

Payload {userName(M), password(M)}

Response {success, sessionToken, <parentData JSON object>}

8. loginParentByUID (POST)

Header {orgUID}

Payload {authenticatedUserUID(M)}

Response {success, sessionToken, <parentData JSON object>}

9. loginChildByUID (POST)

Header {orgUID}

Payload {authenticatedUserUID(M)}

Response {success, sessionToken, <childData JSON object>}

10. getChildVerificationCode (GET)

Header {orgUID, sessionToken}

Payload {childId(M)}

Response {success, verificationCode}

11. authenticateChildVerificationCode (POST)

Header {orgUID}

Payload {verificationCode(M)}

Response {success, childId, authenticatedUserUID}

12. getChildrenByParentId (GET)

Header {orgUID, sessionToken}

Payload {parentId(M)}

Response {success, childData [<array of childData JSON objects>]}

13. addBookToParentBookShelf (POST)

Header {orgUID, sessionToken}

Payload {bookId(M), parentId(M)}

Response {success}

14. getBookShelfByParentId (GET)

Header {orgUID, sessionToken}

Payload {parentId(M)}

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- Response {success, bookData [<array of parentBook-Data JSON objects>]}
15. getBookShelfByParentIdNew (GET)
Header {orgUID, sessionToken}
Payload {parentId(M), pageNumber(M), numberOfRe- 5
cords(M)}
Response {success, bookCount, bookData [<array of parentBookData JSON objects>]}
16. addBookToChildBookShelf (POST) 10
Header {orgUID, sessionToken}
Payload {bookId(M), childId(M)}
Response {success}
17. getBookShelfByChildId (GET)
Header {orgUID, sessionToken}
Payload {childId(M)} 15
Response {success, childBookData [<array of child-BookData JSON objects>,<array of assignmentData JSON objects>]}
18. getBookShelfByChildIdNew (GET) 20
Header {orgUID, sessionToken}
Payload {childId(M), pageNumber(M), numberOfRe-
cords(M)}
Response {success, bookCount, childBookData [<ar-
ray of childBookData JSON objects>,<array of 25
assignmentData JSON objects>]}
19. getBookInfoByBookIdAndChildId (GET)
Header {orgUID, sessionToken}
Payload {bookId(M), childId(M)}
Response {success, childBookData [<array of book- 30
Data JSON objects>,<array of assignmentData
JSON objects>]}
20. getValidationMetadataBybookIdAndChapterId
(GET)
Header {orgUID, sessionToken} 35
Payload {bookId(M), chapterId(M)}
Response {success, validationMetadata}
21. updateBookAssignmentForChild (POST)
Header {orgUID, sessionToken}
Payload {childId(M), bookId(M), dueDate(O), assign- 40
mentDate(O), readingStartDate(O), unit(O), numU-
nits(O), freqInDays(O), deliveryInterval (O), deliv-
eryStartTime (O), deliveryEndTime (O)}
Response {success}
22. updateBookTrackingForChild (POST)/*if hasCom- 45
pleted is true then completionDate is mandatory*/
Header {orgUID, sessionToken}
Payload {childId(M), bookId(M), readTime(O), read-
Percentage(O), score(O), accuracyIndex, chapter-
Counter(O), spoonfulCounter(O), wordCounter(O), 50
questionCounter(O), hasCompleted(O), completion-
Date(O)}
Response {success}
23. pointsRedeemedByChildId (POST)
Header {orgUID, sessionToken} 55
Payload {childId(M), rewardId(M), pointsValue(M)}
Response {success}
24. getChildrenByBookId (GET)
Header {orgUID, sessionToken}
Payload {bookId(M), parentId(M)} 60
Response {success, childData [<array of childData
JSON objects>]}
25. getRewardList (GET)
Header {orgUID, sessionToken}
Payload { } 65
Response {success, rewardData[array of rewardData
JSON objects]}

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26. getRewardListNew (GET)
Header {orgUID, sessionToken}
Payload {pageNumber(M), numberOfRecords(M)}
Response {success, rewardCount, rewardData[array of
rewardData JSON objects]}
27. getRewardsByChildId (GET)
Header {orgUID, sessionToken}
Payload {childId(M)}
Response {success, childRewardData [<array of child-
CurrentRewardData JSON objects>,<array of
rewardData JSON objects>]}
28. getRewardCategories (GET)
Header {orgUID, sessionToken}
Payload { }
Response {success, rewardCategoryData[array of
rewardCategoryData JSON objects]}
29. getRewardsByCategoryId (GET)
Header {orgUID, sessionToken}
Payload {rewardCategoryId(M)}
Response {success, rewardData[array of rewardData
JSON objects]}
30. getRewardsByCategoryIdNew (GET)
Header {orgUID, sessionToken}
Payload {rewardCategoryId(M), pageNumber(M),
numberOfRecords(M)}
Response {success, rewardCount, rewardData[array of
rewardData JSON objects]}
31. addRewardsToParentList (POST)/*rewardIds is an
array of reward id's, Ex:-[1,2,3]*/
Header {orgUID, sessionToken}
Payload {parentId(M), rewardIds(M)}
Response {success}
32. removeRewardsFromParentList (POST)/*rewardIds
is an array of reward id's, Ex:-[1,2,3]*/
Header {orgUID, sessionToken}
Payload {parentId(M), rewardIds(M)}
Response {success}
33. addRewardToChildList (POST)
Header {orgUID, sessionToken}
Payload {childId(M), rewardId(M)}
Response {success}
34. removeRewardsFromChildList (POST)/*rewardIds is
an array of reward id's, Ex :-[1,2,3]*/
Header {orgUID, sessionToken}
Payload {childId (M), rewardIds(M)}
Response {success}
35. getRewardsByParentId (GET)
Header {orgUID, sessionToken}
Payload {parentId(M)}
Response {success, parentRewardData[array of paren-
tRewardData JSON objects]}
36. requestReward (POST)
Header {orgUID, sessionToken}
Payload {childId(M), rewardId(M), silentReplace(M)}
Response {success}
37. approveReward (POST)
Header {orgUID, sessionToken}
Payload {childId(M), rewardId(M), pointsNeeded(M)}
Response {success}
38. claimReward (POST)
Header {orgUID, sessionToken}
Payload {childId(M), rewardId(M)}
Response {success}
39. fulfillRewardStarted (POST)
Header {orgUID, sessionToken}
Payload {childId(M), rewardId(M)}
Response {success}

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40. fulfillRewardCompleted (POST)
Header {orgUID, sessionToken}
Payload {childId(M), rewardId(M)}
Response {success}
41. denyReward (POST)
Header {orgUID, sessionToken}
Payload {childId(M), rewardId(M)}
Response {success}
42. getRewardHistoryByChildId (GET)
Header {orgUID, sessionToken}
Payload {childId(M)}
Response {success, rewardHistoryData[array of rewardHistoryData JSON objects]}
43. trackSpoonfulByChild (POST)
Header {orgUID, sessionToken}
Payload {childId(M), bookId(M), chapterId(M), spoonfulId(M), questionId(M), date(M), totalAttempts(M), successfulAttempts(M), readTime(M), pointsValue(M), numUnitsCompleted(M)}
Response {success}
44. getDeliveryDataForChild (GET)
Header {orgUID, sessionToken}
Payload {childId(M), bookId(M), date(M)}
Response {success, numUnitsGoal, numUnitsCompleted}
45. getTrackingDataByChildAndBook (GET)
Header {orgUID, sessionToken}
Payload {childId(M), bookId(M)}
Response {success, totalAttempts, successfulAttempts, readTime, points}
46. updateUserDeviceToken (POST)
Header {orgUID, sessionToken}
Payload {userId(M), id(M), deviceType(M), deviceId(M), deviceToken(M)}
Response {success}
47. createEvent (POST)
Header {orgUID, sessionToken}
Payload {fromId(M), fromUserType(M), toId(M), toUserType(M), eventType(M), message(M), isNotificationRequired(M), objectTypeOne(O), objectIdOne(O), objectTypeTwo(O), objectIdTwo(O), objectTypeThree(O), objectIdThree(O)}
Response {success}
48. getEvents (GET)
Header {orgUID, sessionToken}
Payload {id(M), userType(M)}
Response {success, eventData [array of eventData JSON objects]}
49. addRegisteredDevice (POST)
Header {orgUID, sessionToken}
Payload {parentId(M), childId(O), deviceName(M), deviceId (M)}
Response {success}

FIG. 11 provides a global view of the data model used for establishing, running a game, and maintaining states for the game.

The data model in general includes the parent (supervisor) and child (reader) relationship (1:many) along with the book-assignment data structures for each child (reader) (1:many).

The book-assignment structure keeps track of the criteria selected by the parent when assigning a book to the child: due-date or none, what measure is used to deliver the material (# of chapters or # of spoonfuls per day), the time restriction for the reading (if any), etc.

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The data structures for the progress/tracking of each child for each e-book they are reading can be based on 2 distinct measures:

- (a) the granularity, being a spoonful-ID, tracking their progress against the goal of how much they should have read on a particular date—based on the assignment data.
- (b) for the Validation Quiz associated with each Spoonful, how many Answers the child has attempted for each Spoonful, and in aggregate across the entire Book, and how many correct Answers they have selected across the set of Spoonfuls they have read and answered.

Maintaining the state of where the reader is in the “reading game” for each electronic document is accomplished using a combination of the above data structures.

FIG. 3 is a simplified illustration of reader procedures executed by a reader application on a smart phone or tablet for example, which may be utilized by a child to traverse the engagement containers of an electronic document, and perform the engagement exercises defined and linked to the engagement container. These procedures communicate with the server using an API, for example as described above, to access, process and update the data structures that support a game based on electronic documents as assigned by the supervisor application.

In this example, the reader procedures include the API interface **200** that enables receiving assignments and reporting status information about assigned electronic documents. The reader procedures include storage **201** in which e-books or other electronic documents are downloaded from the server and held for traversal by the game. The reader procedures include e-book engagement container pacing logic supporting user interface operations for traversing the electronic documents according to parameters of the assignment. Also, logic **203** for tracking and reporting status of the traversal of the electronic documents is provided. The reader application includes tools **205** for displaying an engagement container for review by an individual, and an engagement tool **206** which presents engagement game interface links to engagement containers in the electronic document. Furthermore, the reader application includes login logic **204**, which can enable a reader to log into the system without providing identifying information in a manner that is supported by the supervisor application.

FIG. 4 is a simplified illustration of supervisor procedures executed by a supervisor application, which can also be on a smart phone or tablet for example, and which may be utilized by a parent or teacher to assign electronic documents to readers, specify parameters of a game including engagement parameters for the engagement process for individual readers, track progress of readers under its supervision, and execute electronic commerce transactions to obtain e-books or other electronic documents and to obtain rewards earned by readers under its supervision according to game rules.

In this example, the supervisor procedures include an API interface **210** to communicate with the server using an API, for example as described above, to access, process and update data structures that support the game based on electronic documents assigned to readers under its supervision. Modules of the supervisor application include logic to maintain information about assigned bookshelves **211**, which comprise electronic documents assigned to readers under supervision. The supervisor application includes an account management module **212** for readers under supervision to use for setting up accounts for reader applications, linking to sources of e-books or other electronic documents,

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assigning performance parameters to individual readers and electronic documents, and other management procedures. Also, status tracker and reporter module **213** is included, by which the supervisor application is able to retrieve status and report status to the server, in a way that allows tracking of performance of readers under its supervision. The supervisor application includes a module **216** linking to the server or other resources for the purposes of supporting electronic commerce to obtain electronic documents and rewards in support of the gaming system. The supervisor application includes tools **215** for display of status, and assignment information for readers under supervision. Also, the supervisor procedures include a login module **214**, which supports authentication and authorization procedures necessary for linking to reader applications, and executing electronic commerce transactions.

FIG. **5** illustrates four user interface screens supported by a reader application, used for traversing electronic documents. Screen **500** shows an image presented when a reader selects an e-book which has been assigned to the reader. The screen shows a cover image of the e-book selected, a picture of a reward which has been presented to the reader which can be delivered upon achieving goals in the game defined by the supervisor, and statistics concerning progress and performance in the engagement procedures. In this example, the statistics include a percentage (e.g. 1%) of the e-book which has been completed, by reviewing the contents of an engagement container and executing the engagement exercise defined and linked to the engagement container. Also, statistics include a score which can indicate a percentage of correct answers provided during the engagement exercises, or other information about progress and performance of the game. Statistics shown in this example also include a number of points earned (145) which can be accumulated towards achieving the reward goal (bounty 30,000) associated with the illustrated reward. In addition, the statistics include a number of gems (zero in this example) which have been earned by the reader, where gems can correspond to additional levels of game interaction beyond the engagement with individual engagement containers for example. These gems can be utilized by the system to define various aspects of gameplay, or to enable different categories of rewards for the readers.

Screen **501** shows display of all or part of a “spoonful,” which is the content defined by the engagement container being reviewed at this stage of traversing the engagement containers. Widgets on the screen **501** include a pause widget which a reader can use to indicate that the content is not currently being reviewed. Also, the widgets on the screen **501** include a continue widget, which can be interpreted as an engagement signal by the reader application to enable execution of the engagement exercise. After receiving the engagement signal, the reader application can display screen **502** based on the engagement data linked to the current engagement container. In this example, the engagement data defines a question and multiple choices for an answer to the question. The reader application accepts input indicating a choice of answers from the reader to provide data responsive to the engagement. This data is evaluated for correctness and a score is assigned and points awarded for the current engagement container based on the received data and parameters defined in the engagement metadata. Screen **503** is displayed upon completion of the interaction showing points awarded for the current engagement container. After screen **503**, a next engagement container is displayed as the reader continues to traverse the engagement containers of the electronic document.

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FIG. **6** illustrates seven user interface screens supported by a supervisor application in one example. Interface screen **520** displays a “bookshelf” assigned to a specific reader comprising a set of electronic documents identified by cover images, titles and author names for example. Interface screen **521** illustrates avatars of readers under supervision, which in this example are characterized as children. The children are identified by pseudonyms and avatars in this example, avoiding use of personal identifying information in the exchange with the server and other aspects of the system. Interface screen **522** illustrates an ad reward screen, whereby a supervisor can select toys or other products to be identified as a reward to be earned by readers under its supervision.

Interface screen **523** is used to support assigning electronic documents to a child under supervision. In the screen, an individual electronic document is identified with parameters including information about the length of the book, such as word count, reading level associated with the book, and number of questions or other engagement units required for completion of the electronic document, according to the game. Also, an avatar corresponding to the child to which the book is to be assigned is illustrated.

Interface screen **524** shows a screen by which a supervisor application can traverse content of the electronic document, including images.

Interface screen **525** shows a screen generated by monitoring the status of a child under supervision. In this screen, the books assigned, the scores achieved and points awarded to the reader, the amount of time spent reading the document, and the percent completion indicating progress within the document are displayed for the supervisor application. Likewise, interface screen **526** illustrates further information about the child under supervision, indicating the number of e-books being traversed, the number completed, the amount of time spent reading the books, the average score, the points (coins) earned by the child associated with a particular game instance, and a lifetime count of the points earned.

FIGS. **7A** and **7B** are simplified flowcharts of a procedure for traversing an electronic document executed by a reader application for example, with the support of the server and the supervisor application. In this example, the application starts with a reader application login (**700**). Upon opening the app, if it is being executed on a platform different than the corresponding supervisor application, then a one time code is entered (**701**) at the reader platform to securely link the reader application to the server while preserving anonymity of the reader relative to the server. This one time code is generated and delivered under control of the supervisor application, and can have a limited time in which it is valid. If the reader application is being executed on the same platform as the supervisor application, then the applications can communicate directly to add the one time code to the reader application, and enable the reader application.

After successful linking to the server, the reader application can display a table of assigned e-books (**702**). The reader application can then wait for selection of an e-book (**703**). Upon selection, the reader application can retrieve information about the selected e-book from the server, including the content of the book, metadata defining engagement containers and engagement data, and the status of the traversal of the e-book by the reader, and proceed to select a current engagement container based on the retrieved status (**704**). Upon selection of the current engagement container based on parameters retrieved from the server, the spoonful defined by the current engagement container can be displayed (**705**). The reader application can then wait for an engagement signal indicating that the reader is ready for the

engagement exercise after reviewing the spoonful (706). After accepting the engagement signal, the engagement tool on the user interface is enabled and displayed (707), presenting prompts for the engagement based on parameters defined in the engagement data linked to the current engagement container. In this example, engagement data can define questions and answers that are derived from content in the spoonful. The reader application then waits for input indicating a response to the engagement exercise (708). After accepting the response, the performance of the exercise is evaluated, points and a score are assigned, and the server is notified of progress in the electronic document (709). The procedure then proceeds to FIG. 7B to process a successful response.

In FIG. 7B, processing of a successful response is illustrated (720). After a successful response, the reader application processes assignment data to determine whether to proceed with a next engagement container or take other action. Thus, in this example, the reader application checks assignment parameters (721). The assignment parameters conclude time of day interval (e.g. between 1 PM and 5 PM), during which the supervisor has limited use of the corresponding e-book (722). If the time of day parameter is violated, then the next engagement container is blocked and the reader application can display a notice on its local display (724). If the time of day parameter is okay, then the reader application checks pacing parameters (723). Pacing parameters can identify a number of chapters or paragraphs for example to be reviewed per day according to the assignment. If the next engagement container violates the pacing parameters, then the next engagement container is blocked and the reader application can display a notice on its local display (724). If the pacing parameters are not violated by the next engagement container, then the next engagement container is selected (725). The contents of the next engagement container are displayed on the local display controlled by the reader application (726). The reader application can then wait for an engagement signal indicating that the reader is ready for the engagement exercise after reviewing the spoonful (727). After accepting the engagement signal, the engagement tool on the user interface is enabled and displayed (728), presenting prompts for the engagement based on parameters defined in the engagement data linked to the current engagement container. In this example, engagement data can define questions and answers that are derived from content in the spoonful. The reader application then waits for input indicating a response to the engagement exercise (729). After accepting the response, the performance of the exercise is evaluated, points and a score are assigned, and the server is notified of progress in the electronic document (730). The procedure then proceeds back to block 720, to process a successful response.

FIGS. 8A and 8B are a simplified flowchart of a procedure executed by a supervisor application, for example. In this example, a supervisor application login is executed (800). This procedure can require entry of a user ID and password (801). Upon successful login, a table of supervised readers is displayed (802). The supervisor application can then wait for input selecting an individual reader (803). Upon accepting the input identifying individual reader, the supervisor application can display performance metrics associated with the reader (804). The supervisor application can wait for further input, such as input to initiate adding an e-book to the bookshelf of an individual reader (805). In response to a signal indicating that the supervisor application has initiated adding an e-book, the supervisor application can retrieve and display the contents of a bookshelf or list of the

electronic documents available to the supervisor application for assignment to readers under its supervision (806). The supervisor application can wait for further input, indicating selection of a particular electronic document (807). Upon selection of a particular electronic document, the procedure continues as shown in FIG. 8B.

As illustrated in FIG. 8B, with the assignment of an e-book to a specific reader, performance parameters can be assigned to the individual reader, and the individual e-book, including for example time of day parameters, pace parameters, unit parameters to be applied in the pacing (chapter, paragraph, spoonful) and due date parameters (810). Upon addition of e-book, the supervisor application can notify the server of the addition of the e-book to the reader application bookshelf along with the assignment parameters (811). The supervisor application can then wait for an input indicating that the supervisor wishes to enable a reader application (812). Upon receipt of the signal indicating a wish to enable a reader application, the server generates a one time code and delivers it to the supervisor application. Upon receipt of the one time code from the server (813), the supervisor application determines whether the reader application to be enabled is to be executed on the same platform (e.g. the same mobile phone) as the supervisor application, or is to be executed on a different platform (814). If the reader application is to be executed on the same platform, the one time code can be automatically entered into the reader application login procedure by the supervisor application, and the platform can be given to the reader (815). If the reader application is to be executed on a different platform, then the one time code displayed by the supervisor application so that it can be communicated to the reader for use (816). The one time code can have a time limit during which it can be utilized.

FIG. 9 shows a portion of an e-book expressed in XML markup language with added metadata defining engagement containers (some of the text characters have been removed to make space on the sheet). In this example, the engagement container metadata is embedded within the electronic document.

Markup corresponding to the embedded metadata includes <SpoonreadBody> and </SpoonreadBody>, which identifies boundaries of the electronic document which has embedded metadata as described herein. Metadata associated with the electronic document, as delivered, including metadata defining the beginning of chapters is maintained. For a chapter in this example, a plurality of engagement containers is defined. In the illustrated portion, metadata defining the engagement containers includes <Spoonful id="1"> and </Spoonful>; <Spoonful id="2"> and </Spoonful>; and <Spoonful id="3"> and </Spoonful>; <Spoonful id="4"> and so on.

In the data structures of FIG. 11, this content like that shown in FIG. 9 is carried in the table "tbl_book_chapter" as the data element "chapter_content".

FIG. 10 illustrates a portion of the engagement data and metadata (validation metadata) that is linked to the engagement container shown in the excerpt of the electronic document shown in FIG. 9. Engagement data and metadata is expressed using XML in this example.

This data structure defines engagement data using markup defining a validation container <Validation> and </Validation> for each chapter in the book, and each spoonful within the chapter. Engagement data includes in this example questions associated with each spoonful, where the engagement data is linked to the corresponding engagement containers in the electronic document using the markup of the

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form `<Spoonful id="1">` and `</Spoonful>`. Engagement units in the engagement data are identified in this example by markup of the form `<Question id="1" type="MultipleChoice" answer="3">` and `</Question>`. Questions within the engagement units are identified by the markup `<q>` and `</q>`. Answers presented in multiple-choice form in this example are identified by the markup `<a>` and `</a>`.

In the data structures of FIGS. 11, the engagement data like that shown in FIG. 10 is carried in the table "tbl_book_chapter" as the data element "validation_metadata".

The diagram shown in FIG. 11 illustrates one implementation of data structures maintained by a server application in support of the presentation system described herein. According to this organization of the data structures, each electronic document utilized in the system is identified by a book table `tbl_book`. Likewise, each reader and each supervisor having an account in the system is identified by user table (not shown). User types include a reader identified by a child table `tbl_child`, and a supervisor identified by a parent table `tbl_parent`. The electronic documents in this example comprise one or more chapters represented by respective chapter tables `tbl_chapter` linked to their corresponding book table `tbl_book`. Contents of the table `tbl_chapter` are illustrated for example with respect to FIG. 9 as discussed above. In this data structure, the individual electronic documents represented by a corresponding table `tbl_book` can also be linked to information about the electronic document in one or more genre tables table `tbl_book_genre`.

In this data structure, an organization entity is identified by a table `tbl_organization`. Representative organizations can include schools, publishers, nonprofit organizations and other groups that may want to participate in the gamification of electronic documents as described herein. One organization may be for example a public organization. Electronic documents represented by a table `tbl_book` can be linked to one or more table `tbl_org_lib`. Organizations identified by the table `tbl_organization` are linked to electronic documents via the respective table `tbl_org_lib` of the table `tbl_book` that identifies the electronic document.

Electronic documents that have been assigned to a particular reader identified by a child library table `tbl_child_lib` which is linked in this example to its corresponding organization via the table `tbl_org_lib` table. Likewise, the electronic documents that have been assigned by a supervisor identified by a supervisor library table `tbl_parent_lib`, which is linked in this example to its corresponding organization via the table `tbl_org_lib`.

Supervisor accounts are maintained in this structure using a parent table `tbl_parent` which is linked to one or more child tables `tbl_child`. In this example, authorization to act as a supervisor can be shared with a teacher or tutor utilizing a teacher-parent table `tbl_teacher_parent`, which is in turn links to a teacher or tutor account using a `tbl_teacher` table.

The data structure illustrated in FIG. 11 includes an event table `tbl_event` utilized to hold notifications and other events that are used by the supervisor application, the server application and the reader application which is necessary to execute their respective functions.

Progress and status of individual readers in corresponding electronic documents is tracked using a spoonful tracking table `tbl_spoonful_tracking`, which is linked to the child table `tbl_child` that is in turn linked to the corresponding `tbl_child_lib` which is in turn linked to the corresponding `tbl_org_lib`. The progress and status of individual readers is tracked also in a points activity table `tbl_points_activity`, which in turn links to the `tbl_org_lib` for connection to

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individual books. Also, the `tbl_spoonful_tracking` is linked to the `tbl_org_lib` that is in turn linked to the corresponding `tbl_book`. In addition, progress and status of individual readers in corresponding electronic documents is maintained in a spoonful progress table `tbl_spoonful_progress`.

The data structures maintained by the server can also include the structures used for identifying and assigning rewards for progress in the review of the electronic documents. In this example, one or more organization reward tables `tbl_org_reward` is linked to the organization table `tbl_organization`. The individual `tbl_org_reward` is linked to one or more child reward tables `tbl_child_reward` and one or more parent reward tables `tbl_parent_reward`. Each child table `tbl_child` can be linked to one or more `tbl_child_reward`. Likewise, each parent table `tbl_parent` can be linked to one or more `tbl_parent_reward`. The status and progress of the individual children, represented by the `tbl_spoonful_progress` in this example is also linked to the `tbl_org_reward`. Rewards that are currently assigned to a specific child are indicated by the child current reward table `tbl_child_current_reward`, one or more of which is linked to each `tbl_child`, and one or more of which is linked to each `tbl_org_lib`. The reward history for individual readers can be maintained in a `tbl_child_reward_history` table, one or more of which is linked to the `tbl_org_reward` table, and one or more of which is linked to each reader via the `tbl_child`.

In this manner, an individual child can be tracked using the spoonful progress table to determine when qualification for a particular reward has been satisfied.

Each parent can have supervisor status for multiple readers which can utilize authorized reader applications under supervision, each represented by the data structure `tbl_child`. The data structure `tbl_child` is associated with a bookshelf including one or more entries in a library of electronic documents, each represented by the table `tbl_child_lib`. The electronic documents included in the child bookshelf are represented by the table `tbl_book`. Contents of the book in this embodiment are carried in the data structure `tbl_book_chapter`.

Books available for assignment can be classified by a source organization, which can have control over a variety of parameters associated with the game as indicated by a number of data structures shown. The number of data structures is used to track progress in the child library, including the tables `tbl_points_activity`, `tbl_spoonful_tracking`, `tbl_spoonful_progress`. Reward status is also tracked in the data structures using `tbl_child_current_reward`, `tbl_child_reward_history`, `tbl_parent_reward`, `tbl_child_reward`, and so on.

The excerpts of the API set forth above are utilized for communication between the supervisor application and the server for the purposes of utilization of these data structures. Also, excerpts of the API set forth above are utilized for communication between the reader application and the server for the purposes of utilization of these data structures.

A server application running in communication with corresponding supervisor applications and reader applications executes procedures for maintaining access to a library of electronic documents using, for example, a data structure like that shown in FIG. 11. The server application can include procedures maintaining access to engagement data for electronic documents in the library, engagement data linked to the engagement containers in electronic documents in the library, and engagement data comprising parameters of an engagement based on the segments of content in the linked engagement containers. The server application can include procedures for maintaining supervisor account

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records, the supervisor account records identifying a set of reader accounts under supervision. The server application can include procedures for maintaining reader account records, the reader account records storing identifiers of electronic documents assigned to reader accounts, and indications of progress in the assigned electronic documents utilizing the data structures illustrated in FIG. 11.

FIG. 12 illustrates a representative organization for the `tbl_book` which can be utilized in the configuration of FIG. 11. FIG. 13 illustrates a representative organization for the `tbl_book_chapter`, linked to a `tbl_book` by a `book_id` parameter. Also, as mentioned above, the elements `chapter_Content` in this example carries the content of a chapter of the electronic document such as that shown in FIG. 9. The element `validation_metadata` carries the engagement data and metadata such as that shown in FIG. 10. Other elements shown have descriptive names. FIG. 14 illustrates a representative organization for the table `tbl_child_current_reward`, linked to an individual child by the `child_id` element. FIG. 15 illustrates a representative organization for the table `tbl_child`, which includes a `child_id` element and a `parent_id` element. FIG. 16 illustrates a representative organization for the table `tbl_child_lib`, which includes a `child_id` element, a `child_lib_id` element and an `org_lib_id` element. This table also includes assignment parameters including time of day parameters, start time parameters, completion date parameters, as well as information related to progress of the individual child through specific electronic documents.

FIG. 17 illustrates a representative organization for the table `tbl_spoonful_tracking`. The data elements in this table are descriptive of their function.

FIG. 18 illustrates a representative organization for the table `tbl_spoonful_progress`.

Of course, data structures of this type can be organized in many ways. FIG. 11 is provided for the purposes of illustration of one approach.

The technology disclosed herein can be implemented in the context of any computer-implemented system including a database system, a multi-tenant environment, or a relational database implementation like an Oracle™ compatible database implementation, an IBM DB2 Enterprise Server™ compatible relational database implementation, a MySQL™ or PostgreSQL™ compatible relational database implementation or a Microsoft SQL Server™ compatible relational database implementation or a NoSQL™ non-relational database implementation such as a Vampire™ compatible non-relational database implementation, an Apache Cassandra™ compatible non-relational database implementation, a BigTable™ compatible non-relational database implementation or an HBase™ or DynamoDB™ compatible non-relational database implementation. In addition, the technology disclosed can be implemented using different programming models like MapReduce™, bulk synchronous programming, MPI primitives, etc. or different scalable batch and stream management systems like Apache Storm™, Apache Spark™, Apache Kafka™, Apache Flink™, Truviso™, Amazon Elasticsearch Service™, Amazon Web Services™ (AWS), IBM Info-Sphere™, Borealis™, and Yahoo!S4™.

FIG. 19 is a simplified block diagram of a network node that can be used to implement the server side, supervisor side and reader side applications or other components of the electronic document engagement system described herein. The network node typically includes an operating system executed by a processor subsystem 314 which communicates with a number of peripheral devices via bus subsystem 312. These peripheral devices may include a storage sub-

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system 324, comprising a memory subsystem 326 and a file storage subsystem 328, user interface input devices 322, user interface output devices 320, and a network interface subsystem 316. The input and output devices allow user interaction with network node 202. Network interface subsystem 316 provides an interface to outside network 218 and is coupled via network 218 to other elements in a data processing system. The physical hardware components of network interfaces are sometimes referred to as network interface cards (NICs), although they need not be in the form of cards; for instance they could be in the form of integrated circuits (ICs) and connectors fitted directly onto a motherboard, or in the form of macrocells fabricated on a single integrated circuit chip with other components of the computer system.

User interface input devices 322 may include a keyboard, pointing devices such as a mouse, trackball, touchpad, or graphics tablet, a scanner, a touchscreen incorporated into the display, audio input devices such as voice recognition systems, microphones, and other types of input devices. In general, use of the term “input device” is intended to include all possible types of devices and ways to input information into network node 202 or onto network 218.

User interface output devices 320 may include a display subsystem, a printer, a fax machine, or nonvisual displays such as audio output devices. The display subsystem may include a cathode ray tube (CRT), a flat panel device such as a liquid crystal display (LCD), a projection device, or some other mechanism for creating a visible image. The display subsystem may also provide a nonvisual display such as via audio output devices. In general, use of the term “output device” is intended to include all possible types of devices and ways to output information from network node to the user or to another machine or network node. In particular, an output device of the network node on which the e-book engagement system is implemented, may include a visual output informing a user of action recommendations made by the system, or may include a communication device for communicating action signals.

Storage subsystem 324 stores the basic programming and data constructs that provide the functionality of certain embodiments of the present invention. For example, the various modules implementing the functionality of certain embodiments of the invention may be stored in storage subsystem 324. These software modules are generally executed by processor subsystem 314.

Memory subsystem 326 typically includes a number of memories including a main random access memory (RAM) 330 for storage of instructions and data during program execution and a read-only memory (ROM) 332 in which fixed instructions are stored. File storage subsystem 328 provides persistent storage for program and data files, and may include a hard disk drive, a floppy disk drive along with associated removable media, a CD ROM drive, an optical drive, or removable media cartridges. The databases and modules implementing the functionality of certain embodiments of the invention may have been provided on a computer-readable medium such as one or more CD-ROMs, volatile memory, non-volatile memory, application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), magnetic and optical storage devices such as disk drives, magnetic tape, CDs (compact discs), DVDs (digital versatile discs or digital video discs), or other media capable of storing computer-readable media now known or later developed. The databases and modules implementing the functionality of certain embodiments of the invention may also be stored by file storage subsystem 328. The host

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memory subsystem 326 contains, among other things, computer instructions which, when executed by the processor subsystem 314, cause the computer system to operate or perform functions as described herein. As used herein, processes and software that are said to run in or on “the host,” “the computer” or “the network,” execute on the processor subsystem 314 in response to computer instructions and data in the host memory subsystem 326 including any other local or remote storage for such instructions and data.

Bus subsystem 312 provides a mechanism for letting the various components and subsystems of network node 202 communicate with each other as intended. Although bus subsystem 312 is shown schematically as a single bus, alternative embodiments of the bus subsystem may use multiple busses.

A network node can be of varying types including a personal computer, a portable computer, smart phone, tablet computer, a workstation, a computer terminal, a network computer, a television, a mainframe, a server farm, a widely-distributed set of loosely networked computers, or any other data processing system or user device. Due to the ever-changing nature of computers and networks, the description of network node depicted in FIG. 19 is intended only as a specific example for purposes of illustrating the preferred embodiments of the present invention. Many other configurations of network node are possible having more or less components than the system depicted in FIG. 19.

In some embodiments, in addition, one or more of the server application, the supervisor application, and the reader application can be implemented in the network node as a Software-as-a-Service (SaaS) application, a web-architected application or a cloud-delivered service. Examples of common SaaS applications today include Salesforce.com™, Box™, Dropbox™, Google Apps™, Amazon Web Services AWS™, Microsoft Office 365™, Workday™, Oracle on Demand™, Taleo™, Yammer™, and Concur™. SaaS applications provide functionalities to users that are implemented in the cloud, and that are the target of policies, e.g., logging in, editing user information, updating whitelists, deleting contacts from the contact list, in contrast to the offerings of simple websites and e-commerce sites. Note that a SaaS application can be supported by both web browser clients and application clients that use URL-based APIs (application programming interfaces).

FIGS. 20-31 are briefly described above, and not described again. These user interface images illustrate procedures executed by the editor application, during intake of an electronic document and configuration with engagement container metadata and engagement data as described herein.

A number of flowcharts illustrating logic executed by aspects of the electronic document engagement and presentation system are set forth herein. For all flowcharts herein, it will be appreciated that many of the steps can be combined, performed in parallel or performed in a different sequence without affecting the functions achieved. In some cases, as the reader will appreciate, a re-arrangement of steps will achieve the same results only if certain other changes are made as well. In other cases, as the reader will appreciate, a re-arrangement of steps will achieve the same results only if certain conditions are satisfied. Furthermore, it will be appreciated that the flow charts herein show only steps that are pertinent to an understanding of the invention, and it will be understood that numerous additional steps for accomplishing other functions can be performed before, after and between those shown.

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While the present invention is disclosed by reference to the preferred embodiments and examples detailed above, it is to be understood that these examples are intended in an illustrative rather than in a limiting sense. It is contemplated that modifications and combinations will readily occur to those skilled in the art, which modifications and combinations will be within the spirit of the invention and the scope of the following claims.

What is claimed is:

1. A method performed by a server for a parent account to manage access to and monitor progress through resources in a nicknamed child account for a child and to acquire more resources, the method comprising:

receiving, by the server, a registration request with a nickname other than the child's real name from the parent account to set up the nicknamed child account, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child;

receiving activity data and tracking the activity data for the child and the assigned one or more e-books;

receiving a purchase request, from the nicknamed child account, to buy an e-book from an electronic bookstore; and

notifying the parent account of the purchase request.

2. The method of claim 1, further including:

enabling the nicknamed child account to access a reward responsive to the tracked activity data indicating the child has reached an educational goal defined by the parent.

3. The method of claim 1, further including:

receiving a request to see the tracked activity data of the child from the parent account; and

reporting the tracked activity data of the child to the parent account.

4. The method of claim 1, wherein the child does not have access to unassigned e-books.

5. The method of claim 1, further including:

receiving, from the parent account, one or more schedule parameters; and wherein the authorization of the device to display the content of the one or more e-books is based on the one or more schedule parameters.

6. The method of claim 5, wherein:

the one or more schedule parameters comprises a bedtime interval; and

the display of the content is disallowed based on the bedtime interval.

7. The method of claim 1, further including:

before receiving the e-book request to assign the one or more e-books to the nicknamed child account, providing, to the parent account, a list of e-books that are available for purchase from an electronic bookstore.

8. The method of claim 1, wherein the one or more e-books comprise a game, the game validates the child's activity with the one or more e-books.

9. The method of claim 1, wherein the activity data is maintained apart from the device.

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10. The method of claim 1, wherein:

the one or more e-books comprise containers that contain segments of the content; and
the segments are defined by paragraph boundaries.

11. The method of claim 1, wherein authorization procedures are elevated for the parent account when compared to the nicknamed child account.

12. The method of claim 1, wherein the nickname comprises a pseudonym or an avatar name.

13. A non-transitory computer readable storage medium impressed with computer program instructions of a server for a parent account to manage access to and monitor progress through resources in a nicknamed child account for a child and to acquire more resources, the non-transitory computer readable storage medium including instructions that, when executed on a processor, implement a method comprising:

receiving, by the server, a registration request with a nickname other than the child's real name from the parent account to set up the nicknamed child account, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child;

receiving activity data and tracking the activity data for the child and the assigned one or more e-books;

receiving a purchase request, from the nicknamed child account, to buy an e-book from an electronic bookstore; and

notifying the parent account of the purchase request.

14. A system including a non-transitory computer readable medium and one or more processors coupled to the non-transitory computer readable medium, the non-transitory computer readable medium impressed with computer program instructions of a server for a parent account to manage access to and monitor progress through resources in a nicknamed child account for a child and to acquire more resources, the non-transitory computer readable medium including instructions that, when executed on a processor, implement a method comprising:

receiving, by the server, a registration request with a nickname other than the child's real name from the parent account to set up the nicknamed child account, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child;

receiving activity data and tracking the activity data for the child and the assigned one or more e-books;

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receiving a purchase request, from the nicknamed child account, to buy an e-book from an electronic bookstore; and

notifying the parent account of the purchase request.

15. A method performed by a server for a parent account to manage access to and monitor progress through resources identified in a nicknamed child account, without the server collecting personal identifying information for a child, the method comprising:

receiving, by the server, a registration request from the parent account to set up the nicknamed child account with a nickname other than the child's real name, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child using the nicknamed child account;

receiving activity data and tracking the activity data for the child based on the assigned one or more e-books, wherein the activity data is maintained apart from the device; and

reporting the activity data tracked for the child upon request from the parent account.

16. A non-transitory computer readable storage medium impressed with computer program instructions for a parent account to manage access to and monitor progress through resources identified in a nicknamed child account, without a server collecting personal identifying information for a child, the non-transitory computer readable storage medium including instructions that, when executed on a processor, implement a method comprising:

receiving, by the server, a registration request from the parent account to set up the nicknamed child account with a nickname other than the child's real name, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child using the nicknamed child account;

receiving activity data and tracking the activity data for the child based on the assigned one or more e-books, wherein the activity data is maintained apart from the device; and

reporting the activity data tracked for the child upon request from the parent account.

17. A system including a non-transitory computer readable medium and one or more processors coupled to the non-transitory computer readable medium, the non-transitory computer readable medium impressed with computer program instructions of a server for a parent account to manage access to and monitor progress through resources

identified in a nicknamed child account, without the server collecting personal identifying information for a child, the non-transitory computer readable medium including instructions that, when executed on a processor, implement a method comprising:

receiving, by the server, a registration request from the parent account to set up the nicknamed child account with a nickname other than the child's real name, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child using the nicknamed child account;

receiving activity data and tracking the activity data for the child based on the assigned one or more e-books, wherein the activity data is maintained apart from the device; and

reporting the activity data tracked for the child upon request from the parent account.

18. A method performed by a parent account to manage access to, monitor progress through and provide rewards based on resources identified in a nicknamed child account for a child, the method comprising:

receiving, by a server, a registration request with a nickname other than the child's real name from the parent account to set up the nicknamed child account, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child using the nicknamed child account;

receiving activity data and tracking activity of the child based on the assigned one or more e-books; and

enabling the nicknamed child account to access a reward responsive to the tracked activity indicating the child has reached an educational goal defined by a parent.

19. A non-transitory computer readable storage medium impressed with computer program instructions of a server for a parent account to manage access to, monitor progress through and provide rewards based on resources identified in a nicknamed child account for a child, the non-transitory

computer readable storage medium including instructions that, when executed on a processor, implement a method comprising:

receiving, by the server, a registration request with a nickname other than the child's real name from the parent account to set up the nicknamed child account, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child using the nicknamed child account;

receiving activity data and tracking activity of the child based on the assigned one or more e-books; and

enabling the nicknamed child account to access a reward responsive to the tracked activity indicating the child has reached an educational goal defined by a parent.

20. A system including a non-transitory computer readable medium and one or more processors coupled to the non-transitory computer readable medium, the non-transitory computer readable medium impressed with computer program instructions of a server for a parent account to manage access to, monitor progress through and provide rewards based on resources identified in a nicknamed child account for a child, the non-transitory computer readable medium including instructions that, when executed on a processor, implement a method comprising:

receiving, by the server, a registration request with a nickname other than the child's real name from the parent account to set up the nicknamed child account, wherein the registration request does not include personally identifying information for the child;

enabling resource access, for the child, via the nicknamed child account on a device;

receiving an e-book request from the parent account to assign one or more e-books to the nicknamed child account, wherein the e-book request includes a child account identifier associated with the nicknamed child account;

assigning the one or more e-books to the device, based on the nicknamed child account and authorizing the device to display content of the one or more e-books to the child using the nicknamed child account;

receiving activity data and tracking activity of the child based on the assigned one or more e-books; and

enabling the nicknamed child account to access a reward responsive to the tracked activity indicating the child has reached an educational goal defined by a parent.

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